PheWAS

HUGEN 2073

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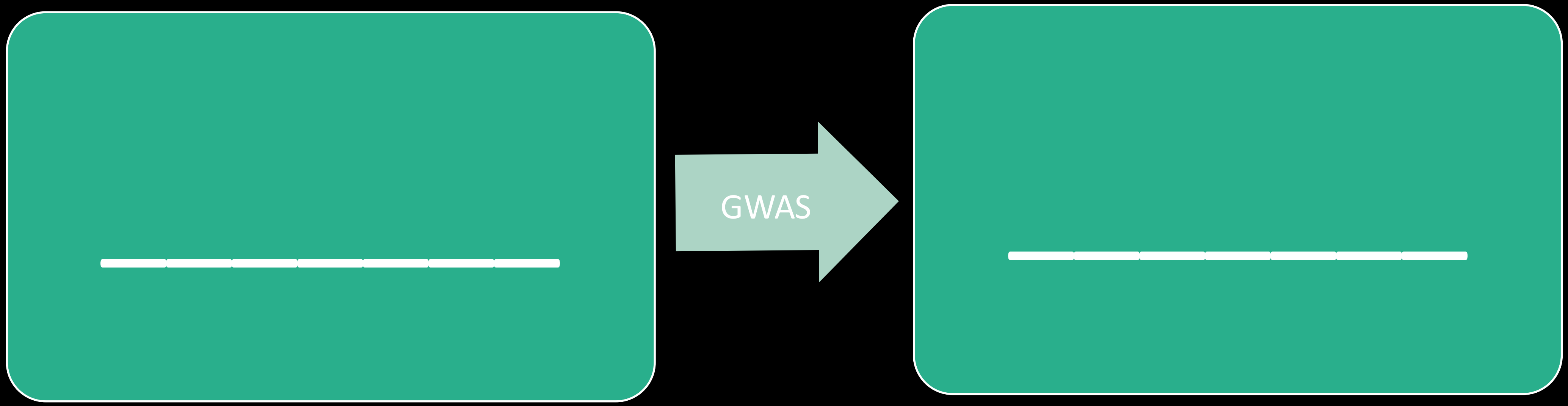
Motivation for pheWAS

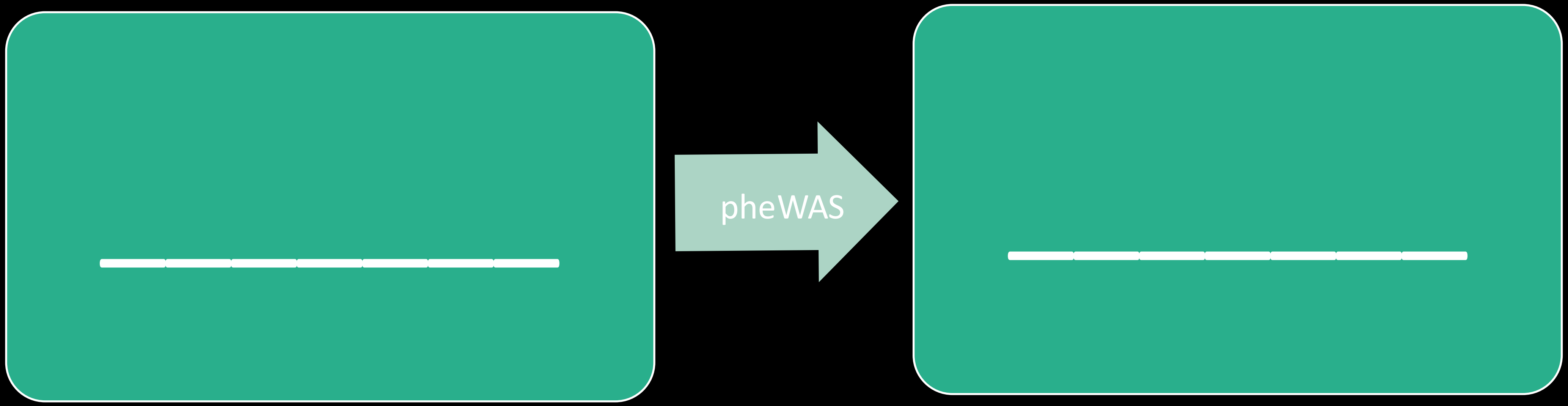
What is the "paradigm" for a GWAS?

What is the direction of inference? Fill in the blanks.
(Hint: think of the terms "forward" and "reverse" genetics.)



Motivation for pheWAS

Can we reverse the GWAS “paradigm?
What would the direction of inference be?
What word goes in the second blank?



Motivation for pheWAS

phenome = set of “all” phenotypes (of an organism, at some level of abstraction)

pheWAS = testing variant(s) for association with phenome

pheWAS asks, “Given a variant (or set of variants), what are all the phenotypes associated with it?”

Motivation for pheWAS

Why do pheWAS? (Discuss)

Are there reasons to suspect this could be a good idea?

Are there any promising applications?

Motivation for pheWAS

Why do pheWAS?
Some reasons:

- “almost 5% of single nucleotide polymorphisms (SNPs) and almost 17% of genes or gene regions are associated with more than one distinct outcome or trait” Sivakumaran, S. *et al*. Abundant pleiotropy in human complex diseases and traits. *Am. J. Hum. Genet.* **89**, 607–618 (2011).
- pheWAS lets us study variants like this more systematically
- Feasibility of studying a variety of *clinical* traits (we’ll discuss why)
- Possibility of using longitudinal data (what is longitudinal data?)
- Potential, e.g., for discovering new drug targets/purposes (we’ll discuss this more) Rastegar-Mojarad, M., Ye, Z., Kolesar, J. M., Hebbring, S. J. & Lin, S. M. Opportunities for drug repositioning from phenome-wide association studies. *Nat. Biotechnol.* **33**, 342–345 (2015).
- PheWAS helps interpret GWAS results

Motivation for pheWAS

In pheWAS we're particularly interested in variants influencing more than one trait. (What is that called?!)

(Note that we are *not necessarily* talking about multivariate testing.)

Motivation for pheWAS

Pleiotropy = when a variant affects more than one phenotype

But a phenotype-genotype association can have different meanings.

pheWAS may help us begin to “dissect” pleiotropic associations.

Note that in pheWAS and in GWAS essentially the same kind of statistical test is performed

- GWAS: “Is genotype associated with phenotype?” (x 1,000,000 variants)
- PheWAS: “Is genotype associated with phenotype?” (x 1,000 phenotypes)

Pleiotropy

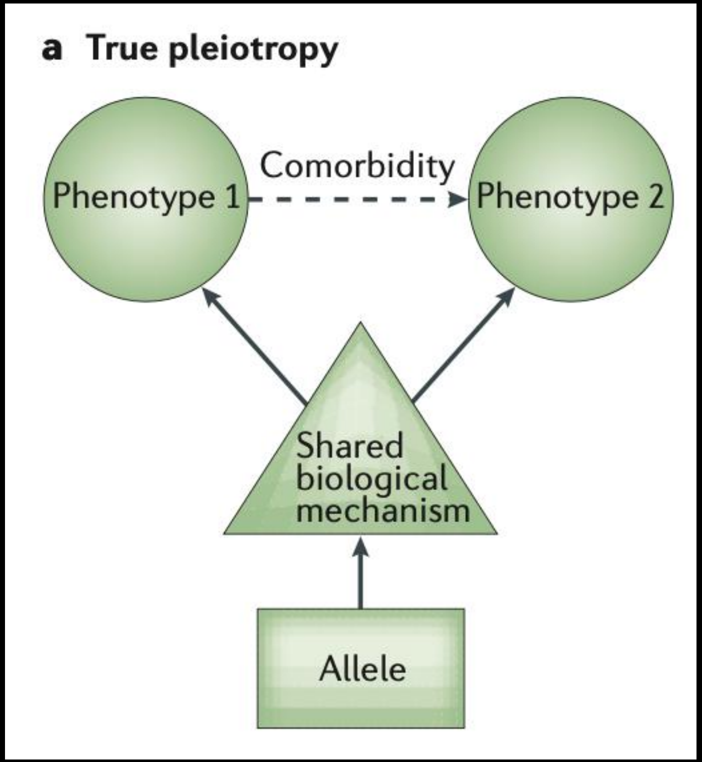
Once you've done a pheWAS, you can announce "rs12345 is associated with traits X and Y" and declare victory, right?

Not quite, but why?

Pleiotropy

Suppose pheWAS tells you rs12345 is associated with phenotype 1 and phenotype 2.

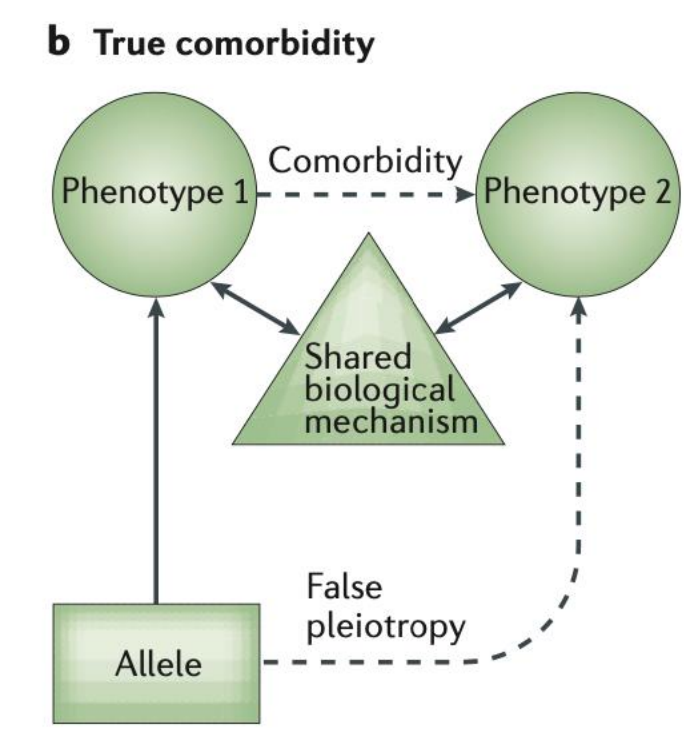
True (or "horizontal") pleiotropy is one possible explanation.



phenotypes 1 and 2 are both <u>independently</u> influenced by the genotypes at a locus influencing a shared biological mechanism.

Pleiotropy

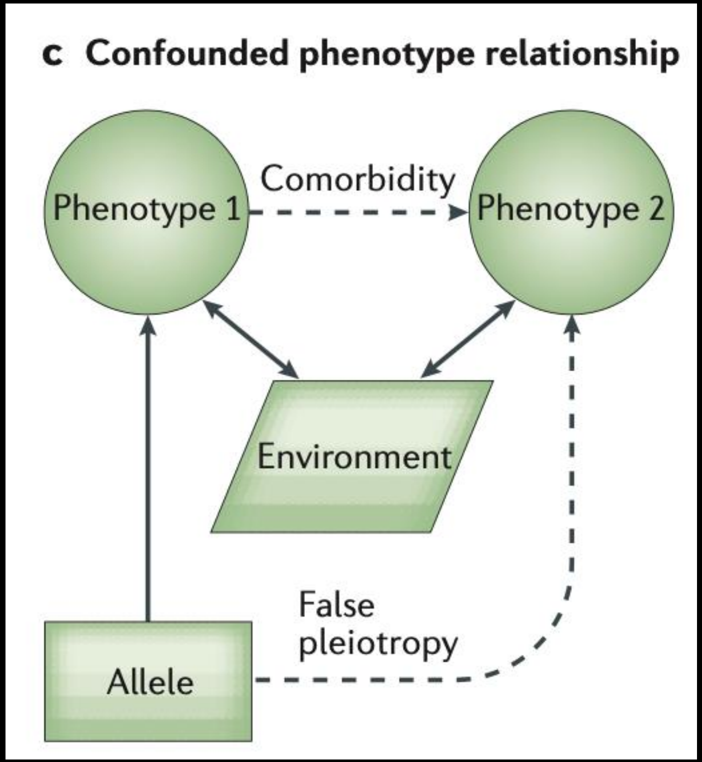
"Comorbidity" could be another explanation.



the phenotypes are causally linked in some way that mediates the effect of the genotype on phenotype 2. In this scenario, the genotypes of the locus influence phenotype 1, which induces phenotype 2, but they <u>do not directly or independently influence phenotype 2</u>

Pleiotropy

Confounding due to shared environmental (or "cryptic") factors could also explain an association between rs12345 and phenotypes 1 and 2.



Non-genetic factors can also modulate the co-occurrence of phenotypes. In this scenario, both phenotypes 1 and 2 have an underlying environmental cause, thus they co-occur in the population. Genotypes at a locus only influence phenotype 1, but <u>both phenotypes 1 and 2 are modulated in similar ways by the environment, and thus seem to be associated</u>

False/vertical pleiotropy

The previous two slides represent false or “vertical” pleiotropy.

The association between genotype and phenotype 2 would disappear if you conditioned on the shared biological/environmental variable.

False/vertical pleiotropy example

Think back to variants in *FTO* being strongly associated with BMI.

They're also associated with T2D.

But when adjusted for BMI, the T2D associations become weaker.

→ Unlikely that *FTO* variants have pleiotropic effects on T2D risk independent of BMI

https://journals.physiology.org/doi/full/10.1152/physiolgenomics.00040.2019?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org

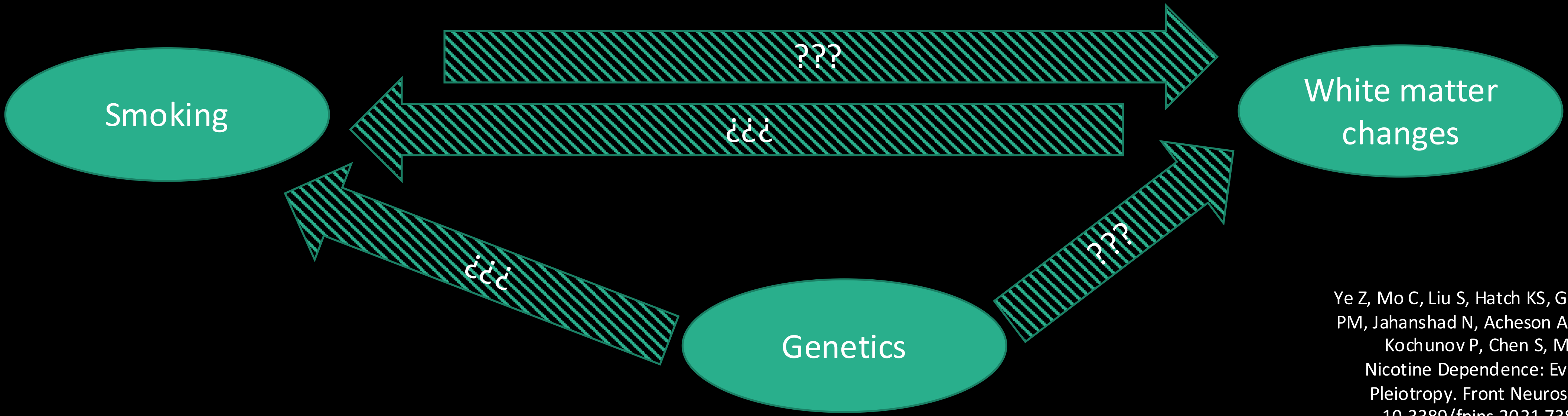
More false pleiotropy examples

Smoking tobacco is addictive and causes cancer and other diseases.

Changes to cerebral white matter (WM) could be related.

So could genetics.

But what causes what?



Ye Z, Mo C, Liu S, Hatch KS, Gao S, Ma Y, Hong LE, Thompson PM, Jahanshad N, Acheson A, Garavan H, Shen L, Nichols TE, Kochunov P, Chen S, Ma T. White Matter Integrity and Nicotine Dependence: Evaluating Vertical and Horizontal Pleiotropy. Front Neurosci. 2021 Oct 14;15:738037. doi: 10.3389/fnins.2021.738037. PMID: 34720862; PMCID: PMC8551454.

More false pleiotropy examples (con’t.)

Here are some possible relationships among the variables…

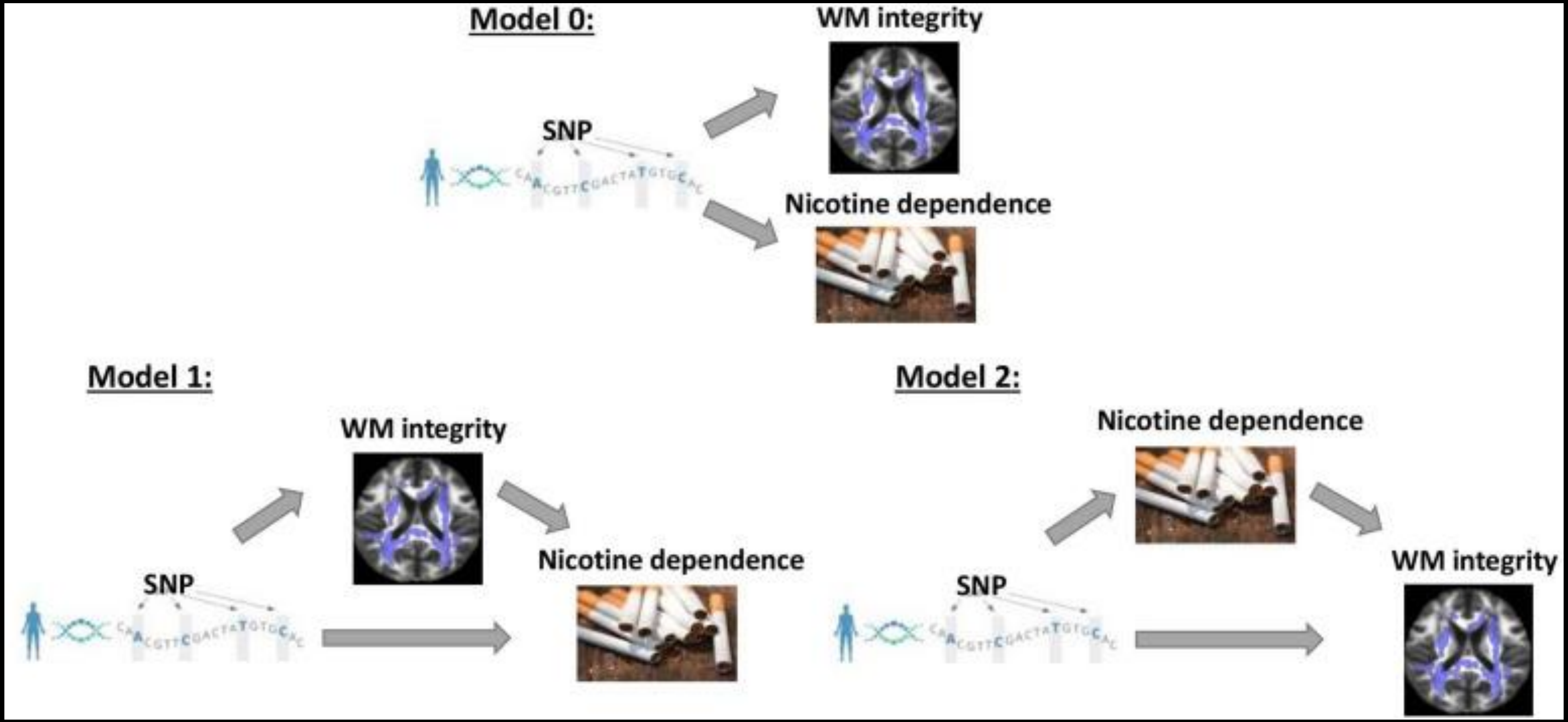
False/vertical pleiotropy:

- Genes → WM integrity → Smoking
- Genes → Smoking → WM integrity

True/horizontal pleiotropy:

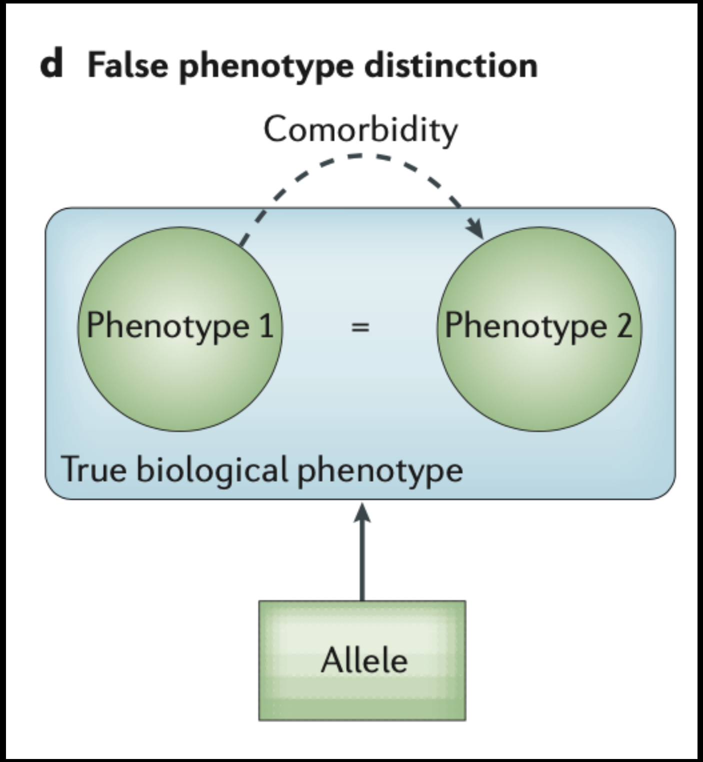
Genes → Smoking & WM integrity (independently)

More false pleiotropy examples (con't.)



Pleiotropy

What if rs12345 is associated with X and Y simply because X and Y are actually the same phenotype? How would this happen?



two ‘phenotypes’ within the phenome represent only <u>one underlying disease</u>

Pleiotropy

PheWAS won’t necessarily disentangle these questions without statistical/functional follow-up!

Interpretation is challenging!

Pleiotropy and pheWAS

Example: rs12203592 (*IRF4*)

- Prior GWAS associations for pigmentation phenotypes
  - Hair color
  - Increased freckling
  - Eye color
  - Increased sunburn
  - Decreased response to tanning
- → good pheWAS SNP
- pheWAS: rs12203592 associated with actinic keratosis (AK), for which skin color is a risk factor

Is AK just a surrogate for skin color?

(If skin color data had been available in the pheWAS, the association could be conditioned on skin color)

An independent study confirmed pigmentation/AK signals for rs12203592 were independent, i.e., remains after adjusting for pigmentation ("true pleiotropy")

phenome

We know what a reference genome is.
So we can reason “trait → variants/genes in genome” (GWAS)

But what is a “reference phenome”?
Unlike the genome, this has to be *defined.*

How is a phenome defined?

There are two main ways for defining phenomes

- Electronic health records (EHRs)
  - Define phenome in terms of billing codes, etc.
- Epidemiological studies
  - Predefine a list of traits to measure/assess

EHR

- Vital signs
- Past/current health conditions
- Diagnostic procedure results (images, etc.)
- Lab results
- Genetics
- Linked to biorepository with genetic data

EHR-based phenomes

Generally use billing/diagnostic/procedural codes from patients' records in repositories.

In the US, ICD-9-CM and ICD-10-CM, and ICD-11 codes have been used.

- Hierarchical system for coding diseases, findings, injuries, procedures, etc.
- Updated sometimes (major source of confusion and difficulty; codes have to be "mapped" between the ICD-9/10/11 for harmonization)
- Tens of thousands of codes

EHR-based phenomes: ICD codes

- List of ICD-9 codes 001–139: infectious and parasitic diseases
- List of ICD-9 codes 140–239: neoplasms
- List of ICD-9 codes 240–279: endocrine, nutritional and metabolic diseases, and immunity disorders
- List of ICD-9 codes 280–289: diseases of the blood and blood-forming organs
- List of ICD-9 codes 290–319: mental disorders
- List of ICD-9 codes 320–389: diseases of the nervous system and sense organs
- List of ICD-9 codes 390–459: diseases of the circulatory system
- List of ICD-9 codes 460–519: diseases of the respiratory system
- List of ICD-9 codes 520–579: diseases of the digestive system
- List of ICD-9 codes 580–629: diseases of the genitourinary system
- List of ICD-9 codes 630–679: complications of pregnancy, childbirth, and the puerperium
- List of ICD-9 codes 680–709: diseases of the skin and subcutaneous tissue
- List of ICD-9 codes 710–739: diseases of the musculoskeletal system and connective tissue
- List of ICD-9 codes 740–759: congenital anomalies
- List of ICD-9 codes 760–779: certain conditions originating in the perinatal period
- List of ICD-9 codes 780–799: symptoms, signs, and ill-defined conditions
- List of ICD-9 codes 800–999: injury and poisoning
- List of ICD-9 codes E and V codes: external causes of injury and supplemental classification

https://en.wikipedia.org/wiki/List_of_ICD-9_codes

Intestinal infectious diseases (001–009) [edit]

- 001 Cholera disease
- 002 Typhoid and paratyphoid fevers
  - 002.0 Typhoid fever
  - 002.1 Paratyphoid fever A
  - 002.2 Paratyphoid fever B
  - 002.3 Paratyphoid fever C
  - 002.9 Paratyphoid fever unspecified
- 003 Other Salmonella infections
  - 003.0 Salmonella gastroenteritis
- 004 Shigellosis
  - 004.9 Shigellosis, unspec.
- 005 Other poisoning (bacterial)
  - 005.0 Staphylococcal food poisoning
- 006 Amoebiasis
  - 006.0 Acute amoebic dysentery without mention of abscess
  - 006.1 Chronic intestinal amoebiasis without mention of abscess
  - 006.2 Amoebic nondysenteric colitis
  - 006.3 Amoebic liver abscess
  - 006.4 Amoebic lung abscess
  - 006.5 Amoebic brain abscess
  - 006.6 Amoebic skin ulceration
  - 006.8 Amoebic infection of other sites
  - 006.9 Amoebiasis, unspecified
- 007 Other protozoal intestinal diseases
  - 007.0 Balantidiasis
  - 007.1 Giardiasis
  - 007.2 Coccidiosis
  - 007.3 Intestinal trichomoniasis
  - 007.4 Cryptosporidiosis
  - 007.5 Cyclosporiasis
  - 007.9 Unspecified protozoal intestinal disease
- 008 Intestinal infections due to other organisms
  - 008.61 Enteritis due to Rotavirus
  - 008.69 Enteritis due to other viral enteritis
  - 008.8 Intestinal infection due to other organism not elsewhere classified
- 009 Ill-defined intestinal infections
  - 009.1 Colitis enteritis and Gastroenteritis of presumed infectious origin

Tuberculosis (010–018) [edit]

- 010 Primary tuberculous infection
  - 010.0 Primary tuberculous infection
  - 010.1 Tuberculous pleurisy in primary progressive tuberculosis
  - 010.8 Other primary progressive tuberculosis
  - 010.9 Primary Tuberculous infection, Unspecifed
- 011 Pulmonary tuberculosis
  - 011.0 Tuberculosis of lung, Infiltrative
  - 011.1 Tuberculosis of lung, Nodular
  - 011.2 Tuberculosis of lung, with Cavitation
  - 011.3 Tuberculosis of bronchus
  - 011.4 Tuberculous fibrosis of lung
  - 011.5 Tuberculous bronchiectasis
  - 011.6 Tuberculous pneumonia
    - any form
  - 011.7 Tuberculous pneumothorax
  - 011.8 Other specified pulmonary tuberculosis
  - 011.9 Pulmonary tuberculosis, Unspecified
    - Respiratory tuberculosis
    - Tuberculosis of lung
- 012 Other respiratory tuberculosis
- 013 Tuberculosis of meninges and central nervous system
- 014 Tuberculosis of intestines, peritoneum, and mesenteric glands
- 015 Tuberculosis of bones and joints
  - 015.0 Tuberculosis of Vertebral column
    - Pott's disease
- 016 Tuberculosis of genitourinary system
- 017 Tuberculosis of other organs
  - 017.1 Erythema nodosum with hypersensitivity reaction in tuberculosis
    - Bazin disease
  - 017.2 Tuberculosis of peripheral lymph nodes
    - Scrofula
- 018 Miliary tuberculosis

Zoonotic bacterial diseases (020–027) [edit]

- 020 Plague
  - 020.0 Bubonic plague
- 021 Tularemia
- 022 Anthrax
- 023 Brucellosis
- 024 Glanders
- 025 Melioidosis
- 026 Rat-bite fever
- 027 Other zoonotic bacterial diseases
  - 027.0 Listeriosis
  - 027.1 Erysipelothrix infection
  - 027.2 Pasteurellosis

- 018 Miliary tuberculosis

Zoonotic bacterial diseases (020–027) [edit]

- 020 Plague
  - 020.0 Bubonic plague
- 021 Tularemia
- 022 Anthrax
- 023 Brucellosis
- 024 Glanders
- 025 Melioidosis
- 026 Rat-bite fever
- 027 Other zoonotic bacterial diseases
  - 027.0 Listeriosis
  - 027.1 Erysipelothrix infection
  - 027.2 Pasteurellosis

Other bacterial diseases (030–041) [edit]

- 030 Leprosy
- 031 Diseases due to other mycobacteria
- 032 Diphtheria
- 033 Whooping cough
- 034 Streptococcal sore throat and scarlatina
  - 034.0 Strep throat
  - 034.1 Scarlet fever
- 035 Erysipelas
- 036 Meningococcal meningitis
- 037 Tetanus
- 038 Septicaemia
  - 038.2 Pneumococcal septicemia
  - 038.4 Septicemia, gram-negative, unspec.
  - 038.9 Septicemia, NOS
- 039 Actinomycotic infections
- 040 Other bacterial diseases
- 041 Bacterial infection in conditions classified elsewhere

Human immunodeficiency virus (HIV) infection (042–044) [edit]

- 042 Human immunodeficiency virus infection with specified conditions
- 043 Human immunodeficiency virus infection causing other specified
- 044 Other human immunodeficiency virus infection

Poliomyelitis and other non-arthropod-borne viral diseases of central nervous system (045–049) [edit]

- 045 Acute poliomyelitis
- 046 Slow virus infection of central nervous system
  - 046.0 kuru
  - 046.1 Creutzfeldt–Jakob disease
- 047 Meningitis due to enterovirus
- 048 Other enterovirus diseases of central nervous system
- 049 Other non-arthropod-borne viral diseases of central nervous system

EHR-based phenomes: ICD codes

Smallpox in ICD-9

ICD9Data.com

Search

Home > 2015 ICD-9-CM Diagnosis Codes > Infectious And Parasitic Diseases 001-139 > Viral Diseases Accompanied By Exanthem 050-059 > Smallpox 050-

2015 ICD-9-CM Diagnosis Code 050.9

Smallpox, unspecified

2015 | Billable Thru Sept 30/2015 | Non-Billable On/After Oct 1/2015

- ICD-9-CM 050.9 is a billable medical code that can be used to indicate a diagnosis on a reimbursement claim, however, 050.9 should only be used for claims with a date of service on or before September 30, 2015. For claims with a date of service on or after October 1, 2015, use an equivalent ICD-10-CM code (or codes).

Convert to ICD-10-CM: 050.9 converts approximately to:
- 2015/16 ICD-10-CM B03 Smallpox

Approximate Synonyms
- Smallpox

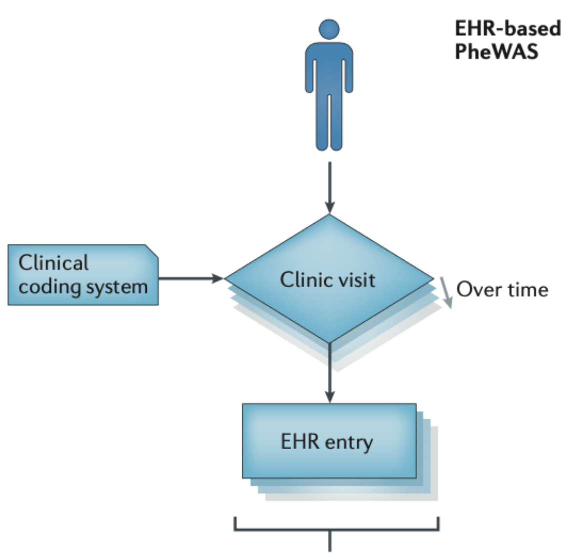
ICD-9-CM Volume 2 Index entries containing back-references to 050.9:
- Infection, infected, infective (opportunistic) 136.9
  - variola 050.9
    - major 050.0
    - minor 050.1
- Smallpox 050.9
  - contact V01.3
  - exposure to V01.3
  - hemorrhagic (pustular) 050.0
  - malignant 050.0
  - modified 050.2
  - vaccination
    - complications - see Complications, vaccination
    - prophylactic (against) V04.1
- Variola 050.9
  - hemorrhagic (pustular) 050.0
  - major 050.0
  - minor 050.1
  - modified 050.2

Pulmonary Tuberculosis ICD-9 codes (incomplete list)

Pulmonary tuberculosis 011- >

- 011 Pulmonary tuberculosis
  - 011.0 Tuberculosis of lung infiltrative
    - 011.00 Tuberculosis of lung, infiltrative, unspecified convert 011.00 to ICD-10-CM
    - 011.01 Tuberculosis of lung, infiltrative, bacteriological or histological examination not done convert 011.01 to ICD-10-CM
    - 011.02 Tuberculosis of lung, infiltrative, bacteriological or histological examination unknown (at present) convert 011.02 to ICD-10-CM
    - 011.03 Tuberculosis of lung, infiltrative, tubercle bacilli found (in sputum) by microscopy convert 011.03 to ICD-10-CM
    - 011.04 Tuberculosis of lung, infiltrative, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.04 to ICD-10-CM
    - 011.05 Tuberculosis of lung, infiltrative, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.05 to ICD-10-CM
    - 011.06 Tuberculosis of lung, infiltrative, tubercle bacilli not found bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.06 to ICD-10-CM
  - 011.1 Tuberculosis of lung nodular
    - 011.10 Tuberculosis of lung, nodular, unspecified convert 011.10 to ICD-10-CM
    - 011.11 Tuberculosis of lung, nodular, bacteriological or histological examination not done convert 011.11 to ICD-10-CM
    - 011.12 Tuberculosis of lung, nodular, bacteriological or histological examination unknown (at present) convert 011.12 to ICD-10-CM
    - 011.13 Tuberculosis of lung, nodular, tubercle bacilli found (in sputum) by microscopy convert 011.13 to ICD-10-CM
    - 011.14 Tuberculosis of lung, nodular, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.14 to ICD-10-CM
    - 011.15 Tuberculosis of lung, nodular, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.15 to ICD-10-CM
    - 011.16 Tuberculosis of lung, nodular, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.16 to ICD-10-CM
  - 011.2 Tuberculosis of lung with cavitation
    - 011.20 Tuberculosis of lung with cavitation, unspecified convert 011.20 to ICD-10-CM
    - 011.21 Tuberculosis of lung with cavitation, bacteriological or histological examination not done convert 011.21 to ICD-10-CM
    - 011.22 Tuberculosis of lung with cavitation, bacteriological or histological examination unknown (at present) convert 011.22 to ICD-10-CM
    - 011.23 Tuberculosis of lung with cavitation, tubercle bacilli found (in sputum) by microscopy convert 011.23 to ICD-10-CM
    - 011.24 Tuberculosis of lung with cavitation, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.24 to ICD-10-CM
    - 011.25 Tuberculosis of lung with cavitation, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.25 to ICD-10-CM
    - 011.26 Tuberculosis of lung with cavitation, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.26 to ICD-10-CM
  - 011.3 Tuberculosis of bronchus
    - 011.30 Tuberculosis of bronchus, unspecified convert 011.30 to ICD-10-CM
    - 011.31 Tuberculosis of bronchus, bacteriological or histological examination not done convert 011.31 to ICD-10-CM
    - 011.32 Tuberculosis of bronchus, bacteriological or histological examination unknown (at present) convert 011.32 to ICD-10-CM
    - 011.33 Tuberculosis of bronchus, tubercle bacilli found (in sputum) by microscopy convert 011.33 to ICD-10-CM
    - 011.34 Tuberculosis of bronchus, tubercle bacilli not found (in sputum) by microscopy, but found in bacterial culture convert 011.34 to ICD-10-CM
    - 011.35 Tuberculosis of bronchus, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.35 to ICD-10-CM
    - 011.36 Tuberculosis of bronchus, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.36 to ICD-10-CM
  - 011.4 Tuberculous fibrosis of lung
    - 011.40 Tuberculous fibrosis of lung, unspecified convert 011.40 to ICD-10-CM
    - 011.41 Tuberculous fibrosis of lung, bacteriological or histological examination not done convert 011.41 to ICD-10-CM
    - 011.42 Tuberculous fibrosis of lung, bacteriological or histological examination unknown (at present) convert 011.42 to ICD-10-CM
    - 011.43 Tuberculous fibrosis of lung, tubercle bacilli found (in sputum) by microscopy convert 011.43 to ICD-10-CM
    - 011.44 Tuberculous fibrosis of lung, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.44 to ICD-10-CM
    - 011.45 Tuberculous fibrosis of lung, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.45 to ICD-10-CM
    - 011.46 Tuberculous fibrosis of lung, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.46 to ICD-10-CM
  - 011.5 Tuberculous bronchiectasis
    - 011.50 Tuberculous bronchiectasis, unspecified convert 011.50 to ICD-10-CM
    - 011.51 Tuberculous bronchiectasis, bacteriological or histological examination not done convert 011.51 to ICD-10-CM
    - 011.52 Tuberculous bronchiectasis, bacteriological or histological examination unknown (at present) convert 011.52 to ICD-10-CM
    - 011.53 Tuberculous bronchiectasis, tubercle bacilli found (in sputum) by microscopy convert 011.53 to ICD-10-CM
    - 011.54 Tuberculous bronchiectasis, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.54 to ICD-10-CM
    - 011.55 Tuberculous bronchiectasis, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.55 to ICD-10-CM
    - 011.56 Tuberculous bronchiectasis, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.56 to ICD-10-CM
  - 011.6 Tuberculous pneumonia (any form)
    - 011.60 Tuberculous pneumonia [any form], unspecified convert 011.60 to ICD-10-CM
    - 011.61 Tuberculous pneumonia [any form], bacteriological or histological examination not done convert 011.61 to ICD-10-CM
    - 011.62 Tuberculous pneumonia [any form], bacteriological or histological examination unknown (at present) convert 011.62 to ICD-10-CM
    - 011.63 Tuberculous pneumonia [any form], tubercle bacilli found (in sputum) by microscopy convert 011.63 to ICD-10-CM
    - 011.64 Tuberculous pneumonia [any form], tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.64 to ICD-10-CM
    - 011.65 Tuberculous pneumonia [any form], tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.65 to ICD-10-CM
    - 011.66 Tuberculous pneumonia [any form], tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.66 to ICD-10-CM
  - 011.7 Tuberculous pneumothorax
    - 011.70 Tuberculous pneumothorax, unspecified convert 011.70 to ICD-10-CM
    - 011.71 Tuberculous pneumothorax, bacteriological or histological examination not done convert 011.71 to ICD-10-CM
    - 011.72 Tuberculous pneumothorax, bacteriological or histological examination unknown (at present) convert 011.72 to ICD-10-CM
    - 011.73 Tuberculous pneumothorax, tubercle bacilli found (in sputum) by microscopy convert 011.73 to ICD-10-CM
    - 011.74 Tuberculous pneumothorax, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.74 to ICD-10-CM
    - 011.75 Tuberculous pneumothorax, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.75 to ICD-10-CM
    - 011.76 Tuberculous pneumothorax, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.76 to ICD-10-CM
  - 011.8 Other specified pulmonary tuberculosis
    - 011.80 Other specified pulmonary tuberculosis, unspecified convert 011.80 to ICD-10-CM
    - 011.81 Other specified pulmonary tuberculosis, bacteriological or histological examination not done convert 011.81 to ICD-10-CM
    - 011.82 Other specified pulmonary tuberculosis, bacteriological or histological examination unknown (at present) convert 011.82 to ICD-10-CM
    - 011.83 Other specified pulmonary tuberculosis, tubercle bacilli found (in sputum) by microscopy convert 011.83 to ICD-10-CM
    - 011.84 Other specified pulmonary tuberculosis, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.84 to ICD-10-CM
    - 011.85 Other specified pulmonary tuberculosis, tubercle bacilli not found by bacteriological examination, but tuberculosis confirmed histologically convert 011.85 to ICD-10-CM
    - 011.86 Other specified pulmonary tuberculosis, tubercle bacilli not found by bacteriological or histological examination, but tuberculosis confirmed by other methods [inoculation of animals] convert 011.86 to ICD-10-CM
  - 011.9 Unspecified pulmonary tuberculosis
    - 011.90 Pulmonary tuberculosis, unspecified, unspecified convert 011.90 to ICD-10-CM
    - 011.91 Pulmonary tuberculosis, unspecified, bacteriological or histological examination not done convert 011.91 to ICD-10-CM
    - 011.92 Pulmonary tuberculosis, unspecified, bacteriological or histological examination unknown (at present) convert 011.92 to ICD-10-CM
    - 011.93 Pulmonary tuberculosis, unspecified, tubercle bacilli found (in sputum) by microscopy convert 011.93 to ICD-10-CM
    - 011.94 Pulmonary tuberculosis, unspecified, tubercle bacilli not found (in sputum) by microscopy, but found by bacterial culture convert 011.94 to ICD-10-CM

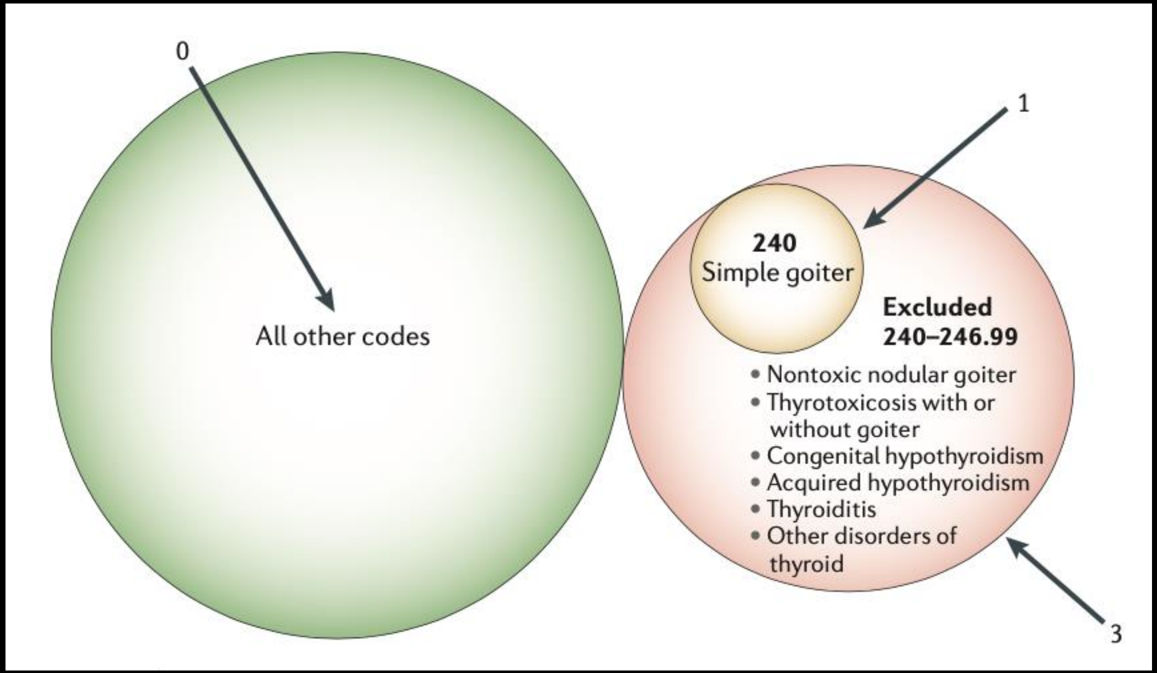
EHR-based phenomes



Patient EHR-based measures

250.0 T2DM	Yes
411.1 coronary syndrome	Yes
414.01 coronary artery disease	No
278.01 obesity	Yes
Alanine aminotransferase	15.6 units per l
Blood albumin	3.7 g per dl
Aspartate aminotrasferase	22 units per l
Bicarbonate (HCO_3)	24 mEq per l
Carbon dioxide (CO_2)	27 mEq per l
Blood cholesterol	240 mg per dl
Blood creatinine	1.2 mg per dl

EHR-based phenomes: defining cases/controls using ICD codes



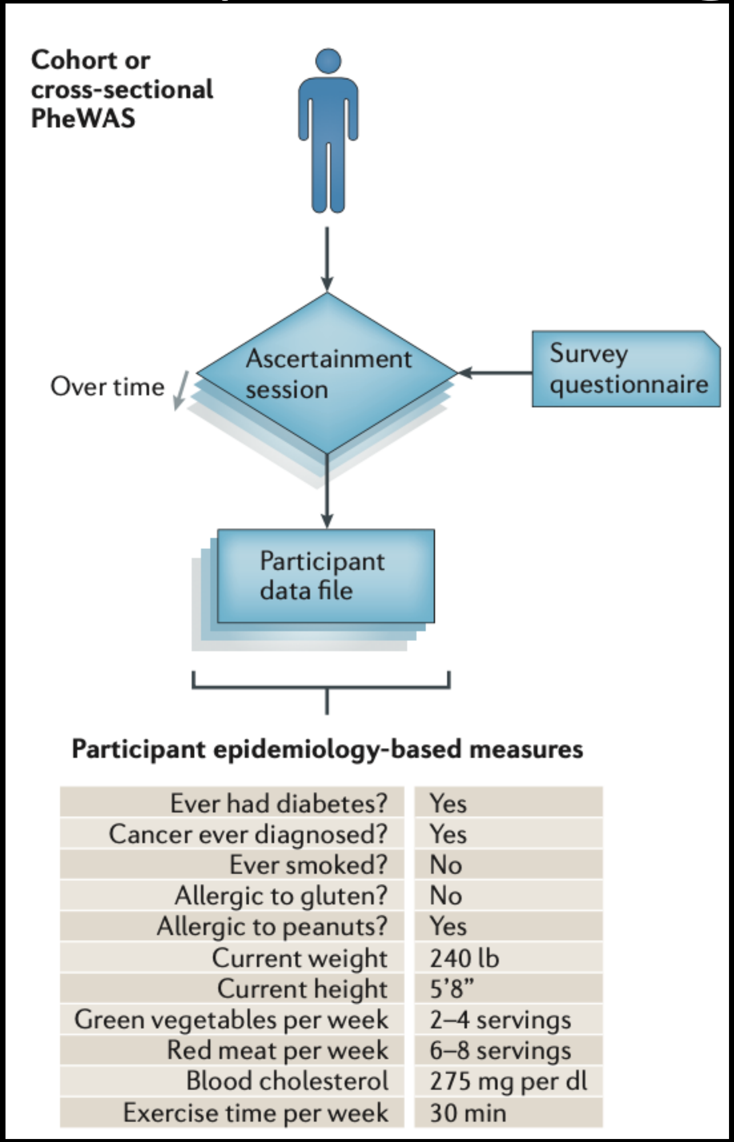
- Cases
  - Decide which code(s)s represent the trait (Phecodes)
  - People with the code(s) are cases
- Controls
  - Decide which codes are *exclusion* criteria
  - Exclude people with those codes
  - Everyone else who isn't a case is a control
  - (Note that a control may not have been assessed for the trait at all!)

EHR-based phenomes: analysis

With cases/controls defined, you can now combine genotype + phenotype information, i.e., run the pheWAS.

Include "age at code" as a covariate. (For controls, use age at last assessment/record).

Epidemiology-based phenomes



Phenotypes collected by

- Exams
- Tests/labs
- Questionnaires

Define cases/controls as desired.

Epidemiology-based phenomes: analysis

A multi-site study may require meta-analysis rather than pooling all the subjects into one analysis. Why? For example,

- Subject consent issues
- Data harmonization issues (protocols/variables may differ across sites)

"Pluses" of meta-analysis

- Ability to explicitly study heterogeneity
- Using discovery/replication cohort design
- (Not so good if some sites have small sample sizes)

Some pheWAS challenges

- Billing codes
  - Mapping between versions
  - “Too many” codes (~500 ICD-9-CM tuberculosis codes)
  - “Too few” codes (T1 and T2 diabetes had the same code)
  - Billing codes aren’t necessarily diagnoses (can reflect simply that a test was ordered, or a condition suspected)
  - One approach for simplification: collapse the decimal part of ICD codes to get to 3-digit codes

More pheWAS challenges

- Algorithmic phenotyping in EHRs
  - Different record types require different algorithms
    - Text (including natural, “uncontrolled” language)
    - Images
    - And so on…
- Epidemiological data
  - Different studies measure/define phenotypes differently
    - PhenX is a system for standardized/portable phenotype/environmental variable definition
  - Somehow these have to be harmonized if they’re to be combined
  - Can be complicated and highly dependent on “what’s available” in each study
  - As in EHRs, collapsing categories is possible
- Multiple testing
  - A lot of tests are performed!
  - How many are independent? What is the correct adjustment?
  - Dimension reduction methods can simply phenotypes and result in fewer tests
- Quality control (data errors, etc.)

Even more pheWAS challenges/concerns

- "clinical order" bias
  - e.g., bone mineral density data may be available almost exclusively for older subjects, causing confounding
  - Epidemiological studies may avoid this
- Environment
  - Environmental data may be "spotty"
  - Epidemiological studies may have more complete/varied data than EHRs
- Real-time and longitudinal data
  - EHRs may have better "real-time" data
  - Both may have longitudinal data
  - Making the best use of repeated measures may not be straightforward

Yet more pheWAS challenges/concerns

- Pharmacogenomics
  - Considered a potentially promising application for pheWAS
  - Records are complicated
    - Requires mining unsystematic, human-generated text
    - Timing/dosage/administration of medication may need to be mined and integrated with a subject's other phenotypes (temporal relationships may be particularly important and difficult to mine)
    - Medication compliance is an extra layer of complexity
- Privacy/consent
- Data access
  - May need to apply to access raw data each time in EHR databases
  - Similarly, dbGaP has genotypes/phenotypes, but each study has to be accessed separately

Even further pheWAS challenges/concerns

Statistical analysis – sometimes multivariate approaches are better (more powerful)

- MultiPhen
- MQFAM (PLINK)
- BIMBAM
- PCHAT
- Others

One advantage: multivariate testing may lower the multiple-testing burden. If a multivariate test is significant for a variant, you might then see which phenotypes drive the multivariate association.

pheWAS example

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OPEN

Phenome-wide association studies across large population cohorts support drug target validation

Dorothée Diogo[1], Chao Tian[2], Christopher S. Franklin[3], Mervi Alanne-Kinnunen[4], Michael March[5], Chris C.A. Spencer[3], Ciara Vangjeli[3], Michael E. Weale[3], Hannele Mattsson[4,6], Elina Kilpeläinen[4], Patrick M.A. Sleiman[5], Dermot F. Reilly[1], Joshua McElwee[1,15], Joseph C. Maranville[1,16], Arnaub K. Chatterjee[1,17], Aman Bhandari[1,18], the 23andMe Research Team, Khanh-Dung H. Nguyen[7], Karol Estrada[7], Mary-Pat Reeve[8], Janna Hutz[8], Nan Bing[9], Sally John[7], Daniel G. MacArthur[10,11], Veikko Salomaa[6], Samuli Ripatti[4,10,12], Hakon Hakonarson[5], Mark J. Daly[10,11], Aarno Palotie[4,10,11,13,14], David A. Hinds[2], Peter Donnelly[3], Caroline S. Fox[1], Aaron G. Day-Williams[1,7], Robert M. Plenge[1,16] & Heiko Runz[1,19]

Motivation

Why pheWAS for drug development?

- “may validate target-disease links”
- may “unravel pleiotropy”, “improve our understanding of the biological functions of a target”, or “hint at concealed pathophysiological connections between disease entities previously considered as distinct”
- “may reveal opportunities for drug repurposing”
- may help discover ADEs (adverse drug events) early
- “may help prioritize multiple possible targets by identifying the target with the most promising therapeutic window”

Motivation

The study asks, “Can pheWAS inform target validation at early stages of drug discovery?”

Methods

pheWAS for 25 SNPs in/near 19 genes

- Genes
  - 9 involved in immune-mediated disease (*ATG16L1, CARD9, CD226, CDHR3, GPR35, GPR65, IFIH1, IRF5, TYK2*)
  - 8 involved in cardiometabolic disease (*F11, F12, GDF15, GUCY1A3, KNG1, LGALS3, PNPLA3, SLC30A8*)
  - 2 involved in neurodegenerative disease (*LRRK2, TMEM175*)
  - Genes were nominated by 25 GWAS SNPs with $p < 5E-8$ for “relevant” phenotypes
  - All either have been shown to influence target gene in functional studies OR to be near a gene “implicated” in a “relevant” mechanism
  - Some poorly understood as candidates, others already targets of drugs in clinical trials

Methods – drug targets investigated

Table 1 Candidate drug targets investigated in the study

	Human genetics		Drug development[b]
Gene	**Prior GWAS associations[a]**	**Mendelian disorders (direction of effect)**	**Indications/status/proposed mechanism of action**
ATG16L1	CD; IBD	–	– / – / –
CARD9	CD; IBD; UC	Familial candidiasis (LOF)	– / – / –
CD226	IBD; MPV; T1D	–	– / – / –
CDHR3	Asthma	–	– / – / –
F11	aPTT; VTE; FXI levels	FXI deficiency (LOF)	Hemophilia C/launched/factor XI stimulant Thrombosis/phase II/factor XI inhibitor
F12	aPTT; FXII levels	Hereditary angioedema (GOF); FXII deficiency (LOF)	Hereditary angioedema; thrombosis/phase I/factor XII inhibitor Antiphospholipid syndrome/preclinical/factor XII inhibitor
GDF15	BMI	–	Cachexia/preclinical/GDF-15 antagonist
GPR35	CD; IBD; UC	–	Cough; mastocytosis; pruritus/phase II/GPR35 agonist
GPR65	CD; IBD; UC	–	– / – / –
GUCY1A3	BP; CAD; MI	Moyamoya 6 with achalasia (LOF)	– / – / –
IFIH1	IgAD; IBD; psoriasis; UC; SLE; T1D; vitiligo	Aicardi–Goutieres syndrome (GOF); Singleton–Merten syndrome (GOF)	Solid cancer/phase I/IFIH1 stimulant (additional targets: RIG-I; TLR3)
IRF5	PBC; RA; SJO; SLE; SSc; UC	–	– / – / –
KNG1	aPTT; FXI levels	–	– / – / –
LGALS3	Galectin-3 levels	–	Liver fibrosis; non-alcoholic steatohepatitis; psoriasis/phase II/galectin-1 and 3 antagonist Pulmonary idiopathic fibrosis/phase II/galectin-3 antagonist Atopic eczema; head and neck cancer; melanoma/phase I/galectin-1 and 3 antagonist Arrhythmia; fibrosis: myocardial, renal; pulmonary hypertension/preclinical/galectin-1 and 3 antagonist Cardiac and renal conditions/preclinical/galectin-3 antagonist
LRRK2	CD; IBD; PD; UC	Familial Parkinson's disease (GOF)	Parkinson's disease/phase I/LRRK2 inhibitor Alzheimer's disease; glaucoma/preclinical/LRRK2 inhibitor
PNPLA3	Alcohol-related cirrhosis; ALT; CT; hepatic steatosis; NAFLD	–	– / – / –
SLC30A8	Fasting glucose; T2D	–	– / – / –
TMEM175	PD	–	– / – / –
TYK2	CD; IBD; MS; PBC; psoriasis; RA; SLE; T1D; UC	Immunodeficiency (LOF)	Atopic eczema/phase II/JAK1 and TYK2 inhibitor psoriasis/phase II/TYK2 inhibitor; JAK1 and TYK2 inhibitor SLE/phase II/TYK2 inhibitor Alopecia areata; UC/phase II/JAK1 and TYK2 inhibitor IBD/phase I/TYK2 inhibitor; JAK1 and TYK2 inhibitor psoriatic arthritis/phase I/TYK2 inhibitor CD/preclinical/JAK1-3 and TYK2 inhibitor cancer: acute leukemia, colorectal, anaplastic large cell lymphoma; MS; RA/preclinical/JAK1 and TYK2 inhibitor Uveitis/preclinical/TYK2 inhibitor

ALT: alanine aminotransferase, aPTT: activated partial thromboplastin time, BMI: body mass index, CAD: coronary artery disease, IgAD: immunoglobulin A deficiency, MI: myocardial infarction, MPV: mean platelet volume, NAFLD: non-alcoholic fatty liver disease, RA: rheumatoid arthritis, SLE: systemic lupus erythematosus, T1D: type 1 diabetes, T2D: type 2 diabetes, VTE: venous thromboembolism, CD: Crohn's disease, IBD: inflammatory bowel disease, MS: multiple sclerosis, PBC: primary biliary cirrhosis, PD: Parkinson's disease, SJO: Sjogren's syndrome, SSc: systemic sclerosis, UC: ulcerative colitis, GOF: gain-of-function, LOF: loss-of-function

[a]Published associations at the genetic locus as defined in Methods. Causal gene not always unambiguously established. For details, see Supplementary Information

[b]As listed in Citeline's Pharmaprojects database. Active development with most advanced status (preclinical or clinical) as of Dec 16, 2017 is indicated

Methods - subjects

pheWAS subjects from 4 cohorts (European ancestry)

- 23andMe (self-reported phenotypes)
- UK Biobank (questionnaire-based health info)
- FINRISK (2 EHR-based subgroups)
- CHOP (Children's Hospital of Philadelphia, EHR-based)

Table 2 Cohorts included in this study

Cohort	Participants geographic distribution	Phenotypes source	*N* binary endpoints tested[a]	Max sample size
23andMe	89% USA (adult)	Questionnaire-based self-reports	654	671,151
Genomics plc UK Biobank	100% UK (adult)	Questionnaire-based self-reports, medical interviews and follow-up	90	112,337
FINRISK	100% Finns (adult)	National health registries (ICD8,9,10)	278	21,371
CHOP	100% USA (pediatric)	Electronic health records (ICD9-CM)	870	12,044
Genomics plc GWAS	Mixed	Mixed—multiple independent disease-specific cohorts	34	-

[a]Number of binary endpoints with N cases ≥ 20

Methods - phenomes

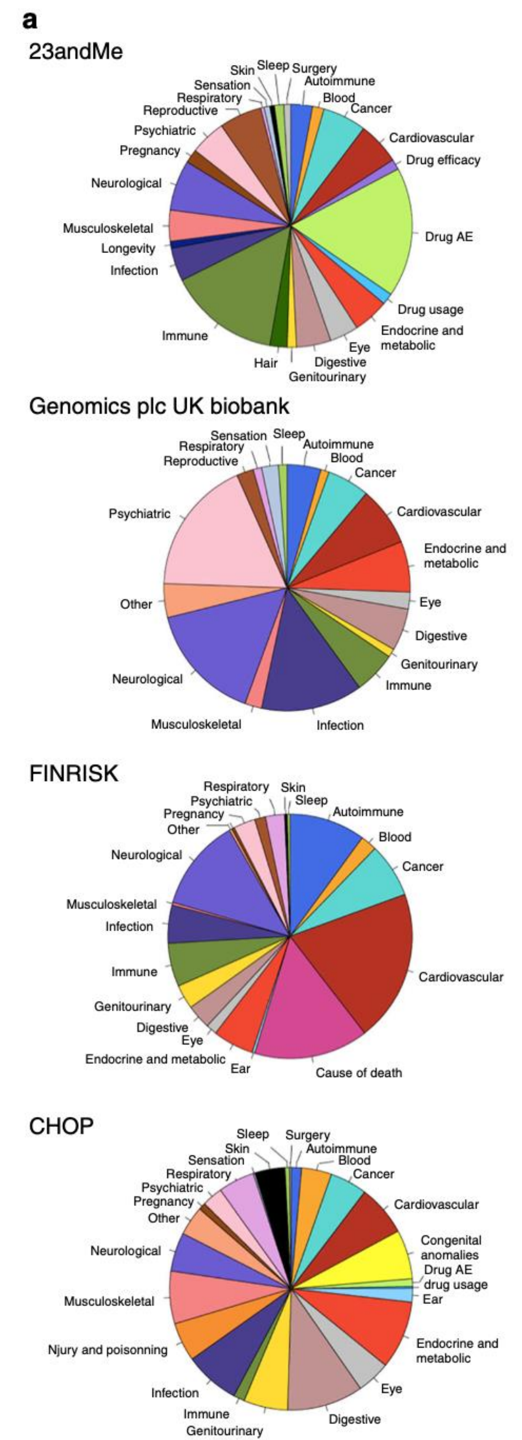
Data formats

- medical interviews
- self-reports
- WHO ICD codes
- ICD9-CM codes

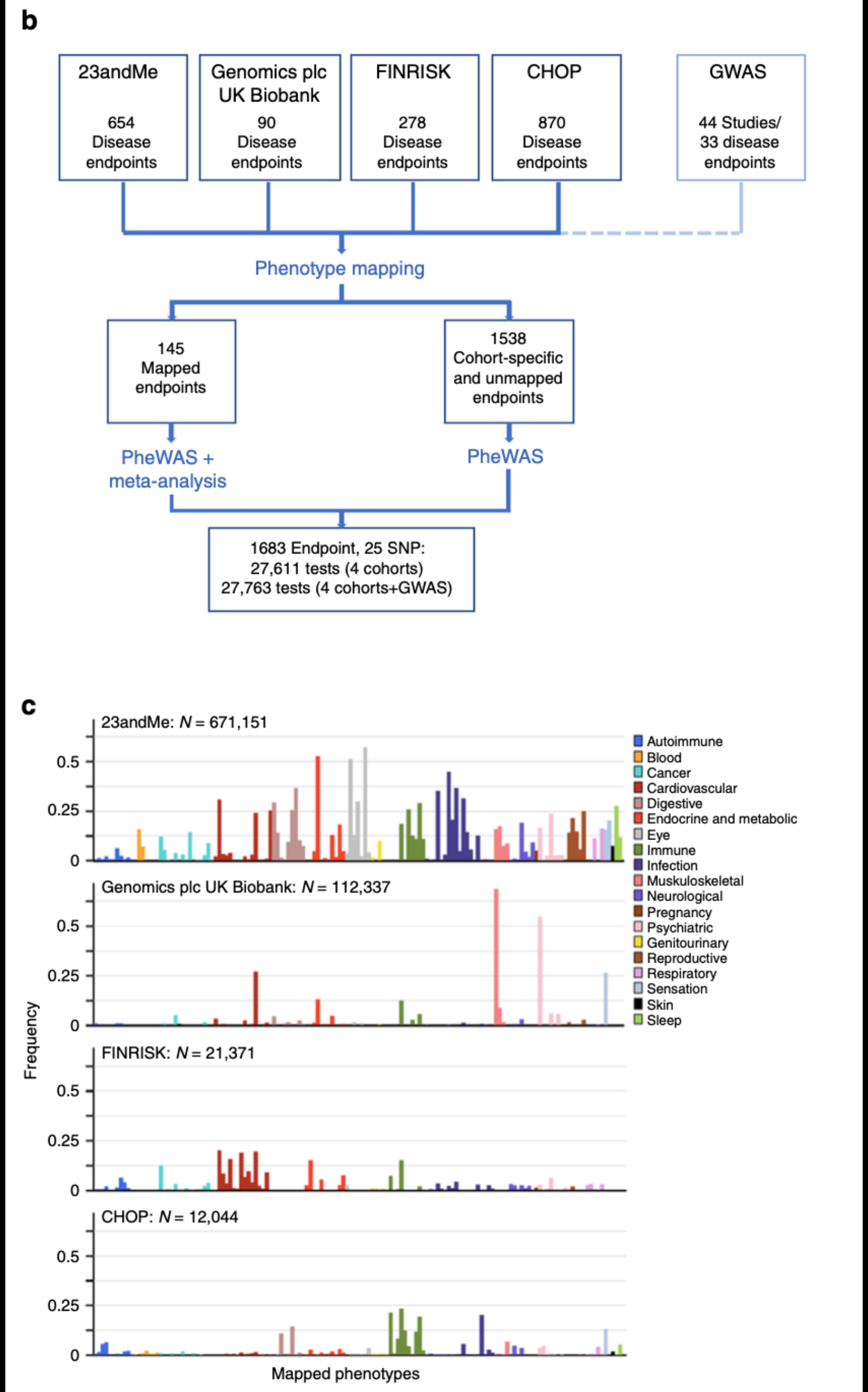
"Manual phenotype mapping identified <u>145 distinct clinical endpoints</u> that were <u>tested in two or more cohorts</u> in <u>up to 697,815 individuals</u>"

1,538 cohort-specific phenotypes (unmapped) were also analyzed.

Phenomes



a
23andMe
Skin
Sleep
Surgery
Autoimmune
Sensation
Respiratory
Blood
Reproductive
Cancer
Psychiatric
Cardiovascular
Pregnancy
Drug efficacy
Neurological
Drug AE
Musculoskeletal
Longevity
Infection
Drug usage
Immune
Endocrine and metabolic
Eye
Hair
Digestive
Genitourinary
Genomics plc UK biobank
Sensation
Sleep
Respiratory
Autoimmune
Reproductive
Blood
Cancer
Psychiatric
Cardiovascular
Endocrine and metabolic
Other
Eye
Digestive
Genitourinary
Neurological
Immune
Musculoskeletal
Infection
FINRISK
Respiratory
Skin
Psychiatric
Sleep
Pregnancy
Autoimmune
Other
Blood
Neurological
Cancer
Musculoskeletal
Infection
Immune
Cardiovascular
Genitourinary
Digestive
Eye
Endocrine and metabolic
Ear
Cause of death
CHOP
Sleep
Surgery
Skin
Autoimmune
Sensation
Blood
Respiratory
Cancer
Psychiatric
Pregnancy
Cardiovascular
Other
Congenital anomalies
Neurological
Drug AE
drug usage
Ear
Musculoskeletal
Njury and poisonning
Endocrine and metabolic
Eye
Infection
Immune
Digestive
Genitourinary



b
23andMe
654 Disease endpoints
Genomics plc UK Biobank
90 Disease endpoints
FINRISK
278 Disease endpoints
CHOP
870 Disease endpoints
GWAS
44 Studies/ 33 disease endpoints
Phenotype mapping
145 Mapped endpoints
1538 Cohort-specific and unmapped endpoints
PheWAS + meta-analysis
PheWAS
1683 Endpoint, 25 SNP:
27,611 tests (4 cohorts)
27,763 tests (4 cohorts+GWAS)
c
23andMe: N = 671,151
Genomics plc UK Biobank: N = 112,337
FINRISK: N = 21,371
CHOP: N = 12,044
0.5
0.25
0
Frequency
Mapped phenotypes
Autoimmune
Blood
Cancer
Cardiovascular
Digestive
Endocrine and metabolic
Eye
Immune
Infection
Muskuloskeletal
Neurological
Pregnancy
Psychiatric
Genitourinary
Reproductive
Respiratory
Sensation
Skin
Sleep

Methods – association testing

Logistic regression + meta-analysis

Statistically, like a GWAS…

…with 25 SNPs instead of 10,000,000

…and 1683 phenotypes instead of 1

Results

9 associations study-wide significant (P < 1.8E−6)
(Where do you think that threshold came from?)

"we identified 71 distinct putative novel associations" (FDR < 0.1 (P < 7 × 10−4))

- 30 mapped
- 41 only in single cohorts

"Forty-three of these putative novel associations showed the same directions of effect as disease endpoints related to the proposed clinical indication for a drug and may hint at potential repositioning opportunities"

Why do you think there are gray cells?

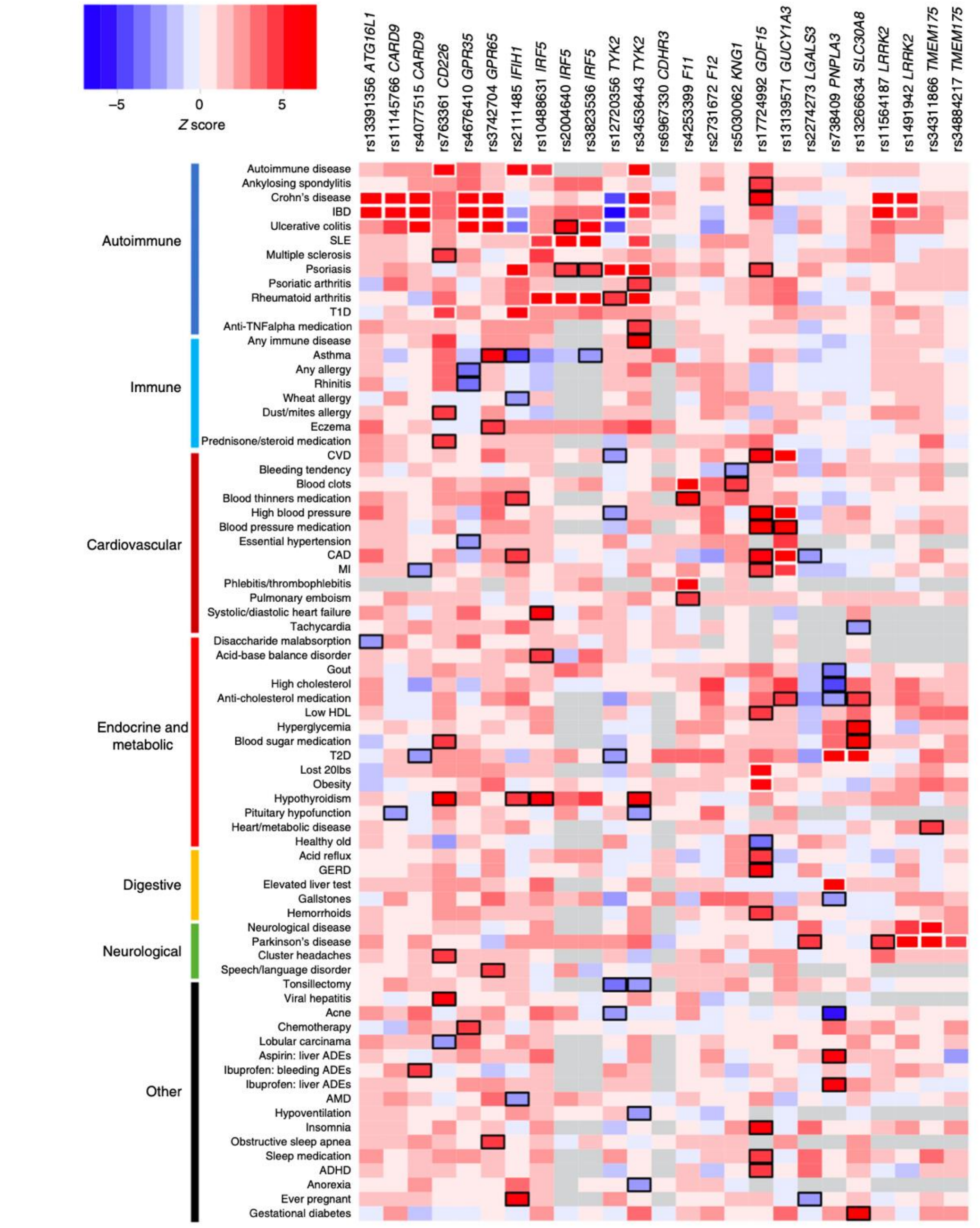


Fig. 2 Meta-PheWAS results for 25 SNPs in candidate drug targets. Phenotypes associated at FDR < 0.1 (*P* < 7e−4) with at least one SNP in the meta-PheWAS are represented. Direction of effect of the known disease-risk increasing allele related to the therapeutic hypothesis is indicated. A positive *Z* score (in red) indicates increased risk, a negative *Z* score (in blue) indicates reduced risk. Known and novel associations reaching FDR < 0.1 are outlined in white and black respectively. Detailed association results are provided in the Supplementary Data 1

Results

9 associations study-wide significant (P < 1.8E−6)

Table 3 Significant novel associations in the PheWAS meta-analysis

				Novel association in meta-PheWAS[a]					
Gene	**SNP**	**EA (EAF)[b]**	**Known associated phenotype[c]**	**Phenotype**	**OR (CI95)**	***P* value**	**Direction[d]**	***N* cases**	***N* controls**
CD226	rs763361	T (0.47)	IBD	Hypothyroidism	1.05 (1.04-1.07)	8.11e−11	++?+?	35,428	412,577
GDF15	rs17724992	A (0.73)	BMI	Heart metabolic disease[e]	1.03 (1.02-1.04)	3.08e−09	+????	275,944	209,302
				High blood pressure[e]	1.03 (1.02-1.04)	7.64e−09	++???	151,511	465,686
				Blood pressure medication[e]	1.03 (1.02-1.04)	1.76e−07	+????	125,406	394,753
				GERD	1.03 (1.02-1.04)	6.11e−07	+????	130,654	384,572
				Any CVD[e]	1.03 (1.01-1.04)	1.40e−06	+????	148,577	388,405
IFIH1	rs1990760	T (0.61)	T1D	**Asthma**[f]	**0.96 (0.95-0.98)**	1.11e−07	− − − − −	57,101	269,659
IRF5	rs10488631	C (0.11)	SLE	Hypothyroidism	1.08 (1.05-1.12)	5.78e−07	++?+?	23,182	236,240
PNPLA3	rs738409	G (0.33)	ALT	**Severe acne**	**0.91 (0.88-0.93)**	1.47e−11	−????	14,812	187,018
				High cholesterol	**0.96 (0.94-0.97)**	1.59e−07	− −???	101,646	180,947
TYK2	rs34536443	G (0.89)	Psoriasis	Any immune disorder	1.10 (1.07-1.13)	4.27e−12	+????	112,148	173,986
				Hypothyroidism	1.14 (1.08-1.20)	1.19e−06	++?−?	23,145	233,757

ALT: alanine aminotransferase, BMI: body mass index, CVD: cardiovascular disease, EA: effect allele, EAF: effect allele frequency, GERD: gastroesophageal reflux disease, IBD: inflammatory bowel disease, SLE: systemic lupus erythematosus, T1D: type diabetes, T2D: type 2 diabetes

[a]Associations reaching $P < 1.8e{-}6$ (Bonferroni-corrected significance threshold) in the meta-analysis of PheWAS results with GWAS results. The full list of potential novel SNP-phenotype pairs reaching FDR < 0.1 is provided in Supplementary Table 5. Novel associations with direction of effect opposite to the known associated disease(s) effect, predicting potential adverse drug events, are highlighted in bold

[b]The effect allele is the risk allele for known associated disease(s) related to the therapeutic hypothesis

[c]Known associated disease related to the therapeutic hypothesis (surrogate for efficacy). The strongest association reported in the literature is indicated. The full list of known associations is provided in Supplementary Table 1

[d]Direction of effect in 23andMe, Genomics plc UK Biobank, FINRISK, CHOP, and GWAS

[e]Correlated phenotypes

[f]Meta-analysis results including the 23andMe, Gplc/UK Biobank, FINRISK, CHOP, and GWAS Gabriel cohorts. When further including the independent GWAS EVE study, the association reaches $P = 6.7 \times 10^{-8}$

Results – replication of novel associations

Meta-analysis of novel associations in another UK BioBank cohort (where possible)

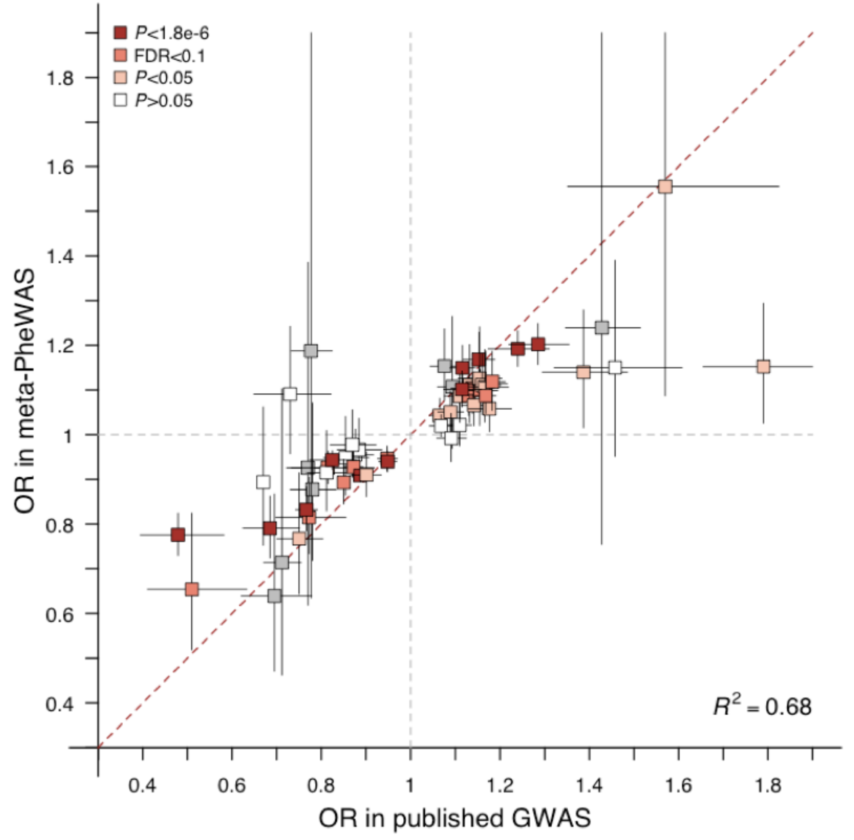
- 41/71 novel associations re-tested
- Includes 8/9 study-wide significant associations

"Overall, meta-analysis with UK Biobank v2 strengthened all eight novel associations with study-wide significance after Bonferroni correction and 23/41 (56%) of the potential novel associations with FDR < 0.1, including eight associations that were based on results from a single PheWAS cohort"

Results – replication of prior GWAS associations

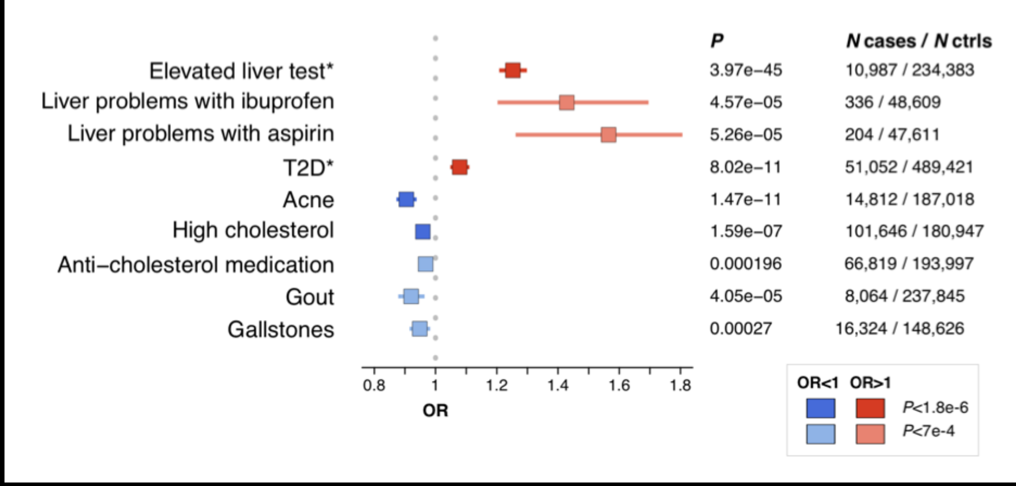
- 25 SNPs had been significant in GWAS for 58 traits
- 47 of the 58 were replicated here ($p < 0.05$)
- (Fewer had fully consistent effect directions)
- Non-replication was “disease-dependent”, possibly the result of ascertainment error for conditions like inflammatory bowel disease, Crohn’s, ulcerative colitis

Results – replication of prior GWAS associations



Supplementary Fig. 2. Effect sizes of known GWAS associations in the meta-PheWAS. Odds ratios (ORs) and 95% confidence intervals in the published GWAS are shown on the x axis. ORs and 95% confidence intervals in the meta-PheWAS are shown on the y axis. Powered associations (beta<0.8, alpha = 0.05) reaching $P<1.8\times10^{-6}$ (Bonferroni corrected pvalue), FDR<0.1 or $P<0.05$ in meta-PheWAS are shown in dark red, red and light red respectively. Powered associations not reaching $P<0.05$ in meta-PheWAS are shown in white. Associations not powered in the meta-PheWAS are shown in grey. The regression coefficient R^2 between published ORs and ORs in meta-PheWAS is indicated.

Results – novel associations



Novel association for PNPLA3 missense SNP rs738409-G (p.I148M)

Previous associations include

- non-alcoholic fatty liver disease (NAFLD), alcohol-related
- cirrhosis and hepatic steatosis
- elevated alanine aminotransferase (ALT) levels

"Consistent findings"

- "elevated liver tests" ($OR_{23andMe}$ = 1.25, $P_{23andMe}$ = 4E−45)
- T2D (OR_{meta} = 1.08, P_{meta} = 8E−11)
- decreased risk for high cholesterol (OR_{meta} = 0.96, P_{meta} = 1.6E−7; Pmeta_v2 = 1.1E−8) and the intake of cholesterol-lowering medications ($OR_{23andMe}$ = 0.97, $P_{23andMe}$ = 2E−4; P_{meta_v2} = 2.8E−5)

Results – novel associations (con't.)

New pheWAS-discovered associations for rs738409-G (missense variant in *PNPLA3*)

- liver toxicities when treated with nonsteroidal anti-inflammatory drugs (NSAIDs)
  - Ibuprofen ($OR_{23andMe}$ = 1.43, $P_{23andMe}$ = 4.6E−5)
  - Aspirin ($OR_{23andMe}$ = 1.57, $P_{23andMe}$ = 5.3E-5)
- acne ($OR_{23andMe}$ = 0.90, $P_{23andMe}$ = 1.5E−11; P_{meta_v2} = 7.3E−12)
- gout (OR_{meta} = 0.92, P_{meta} = 4.1E−5; P_{meta_v2} = 3.9E−9)
- gallstones (OR_{meta} = 0.95, P_{meta} = 2.7E−4; P_{meta_v2} = 1.5E−5)
- associations remained "prominent" after adjusting for elevated liver tests

Known previous associations: G-allele increases liver disease risk, likely through gain of function.

<u>Hypothesis: therapeutic inhibition of *PNPLA3* could treat liver diseases</u>.

"inverse associations with multiple other endpoints, including acne and high plasma cholesterol levels, indicate potential clinically relevant on-target ADEs that should be considered in decisions to progress *PNPLA3* inhibitors toward clinical development"

Results – novel associations (con’t.)

rs1990760-C allele (*IFIH1*, which encodes MDA5)

Previous associations, mostly supported by pheWAS

- Lower risk for
  - type 1 diabetes, T1D
  - vitiligo
  - systemic lupus erythematosus, SLE
  - Psoriasis
- Higher risk for
  - ulcerative colitis

Results – novel associations (con't.)

rs1990760-C allele (*IFIH1*)

Novel associations:

- risk for asthma (OR_{meta} = 1.04, P_{meta} = 6.E 10−8, P_{meta_v2} = 2E−8)
- Significant after adjustment for autoimmune disease (23andMe)
- "Support[s] the hypothesis that <u>inhibition of *MDA5* may protect against several autoimmune diseases</u>. However, our results also reveal the potential of *MDA5* inhibitors to cause pulmonary ADEs and strengthen previous findings for an increased risk for colitis-related symptoms, endpoints that may limit the therapeutic window of *MDA5* modulators and should be considered for monitoring in clinical trials."

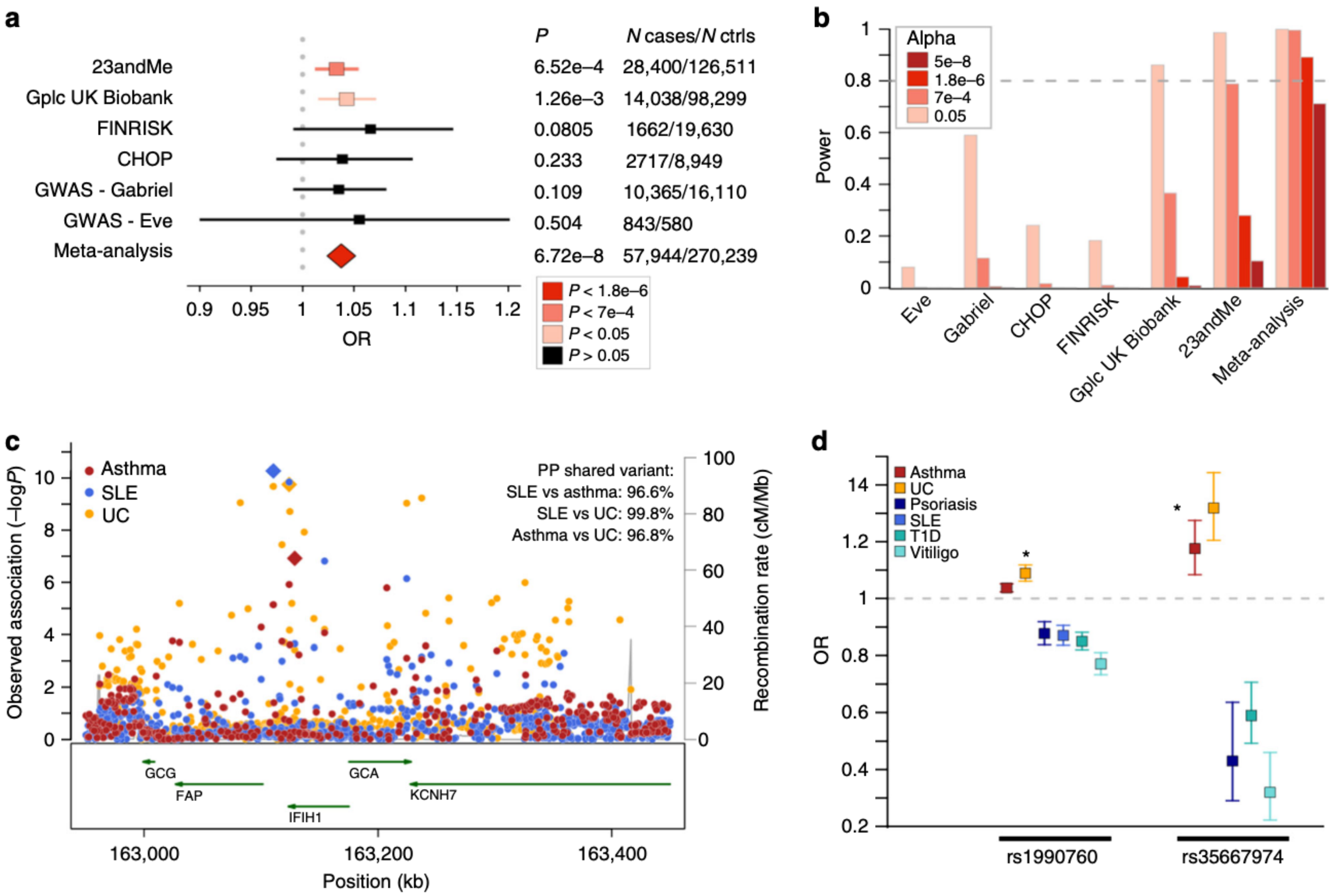


Fig. 3 Pleiotropic effects of *IFIH1* LOF variants. **a** A significant association of *IFIH1* rs1990760-C (p.T946A) with increased risk of asthma was observed in the meta-analysis of PheWAS and GWAS results, with consistent effect estimate across the six cohorts tested. Odds ratios (OR) and 95% confidence intervals are represented. **b** Power estimation demonstrates the lack of power to detect an association at rs1990760-C in currently available asthma GWAS studies. Power to surpass various significance cutoffs ($P < 0.05$; FDR < 0.1, $P < 7e-4$; study-wide significance after Bonferroni correction, $P < 1.8e-6$; and genome-wide significance, $P < 5e-8$) in the six cohorts was estimated using the frequency of the asthma risk allele (RAF = 0.39), the OR in the PheWAS/GWAS meta-analysis (OR = 1.037), a disease prevalence of 8%, and the number of cases and controls in each of the cohorts. **c** Co-localization analysis demonstrates that the asthma, systemic lupus erythematosus (SLE), and ulcerative colitis (UC) associations at the *IFIH1* locus are driven by a shared causal signal. Regional association results with asthma (red), SLE (blue) and UC (orange) are shown. PP, posterior probability of co-localization. **d** Results from this study (indicated by an asterix) combined with previously published findings suggest an allelic series of LOF *IFIH1* alleles decreasing the risk of various autoimmune diseases while increasing the risk of asthma and UC. OR and 95% confidence intervals of association for the *IFIH1* loss-of-function alleles rs1990760-C (p.T946A) and rs35667974-C (p. I923V) are shown

Results – target prioritization for thromboembolism

F11, F12, KNG1 are members of a coagulation pathway.

*"*Anti-coagulation therapies directed against these factors are hypothesized to have improved therapeutic windows over current standard-of-care, which is accompanied by significant bleeding liabilities *"*

Results – target prioritization for thromboembolism

- rs4253399-T
  - reduces circulating FXI levels
  - increases aPTT (blood clotting risk biomarker)
  - PheWAS
    - expected lower risk for blood clots (OR_{meta} = 0.84, P_{meta} = 3.5E−25)
    - no evidence for association with bleeding tendency ($OR_{23andMe}$ = 1.04, $P_{23andMe}$ = 0.35)
- rs5030062-A
  - reduces plasma kininogen and circulating FXI
  - increases aPTT
  - PheWAS
    - Expected lower risk for blood clots (OR_{meta} = 0.93, P_{meta} = 1.6E−4)
    - Expected higher risk for bleeding tendency ($OR_{23andMe}$ = 1.14, $P_{23andMe}$ = 4.1E−4)
- rs2731672- T
  - Reduces FXII levels
  - increases aPTT
  - PheWAS
    - Expected lower risk for blood clots ($OR_{23andMe}$ = 0.96, $P_{23andMe}$ = 0.034)
    - Expected higher risk for bleeding tendency ($OR_{23andMe}$ = 1.09, $P_{23andMe}$ = 0.039)

Results – target prioritization for thromboembolism

"targeting FXI may yield the best compromise between thromboembolism risk reduction and increased bleeding liability, which is consistent with the outcomes of a recent phase 2 clinical trial"

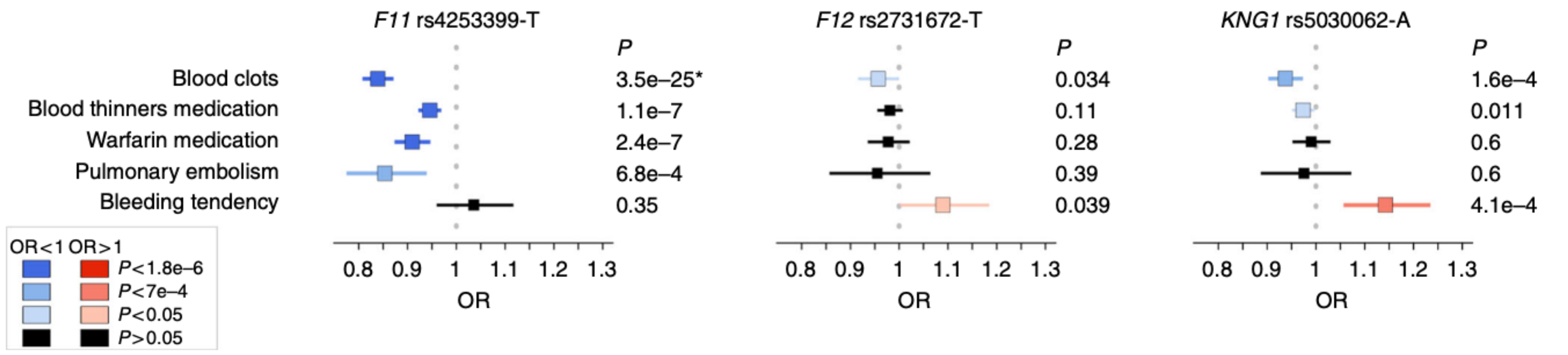


Fig. 4 PheWAS for contact activation coagulation pathway targets. Three SNPs known to affect plasma protein levels of FXI (rs4253399), FXII (rs2731672), and KNG1 (rs5030062), and previously reported associated with partial thromboplastin time (aPTT) were interrogated in meta-PheWAS. Five phenotypes were observed as significantly associated (FDR < 0.1) with at least one of the three SNPs: blood clots (23andMe, FINRISK, and CHOP: 7487 cases, 273,305 controls), known association with the *F11* SNP (*), blood thinners medication (23andMe: 22,985 cases, 236,431 controls), warfarin medication (23andMe: 7142 cases, 94,701 controls), pulmonary embolism (Gplc/UK Biobank: 949 cases, 111,077 controls), and bleeding tendency (23andMe: 1574 cases, 85,223 controls). Odds ratios (OR) and 95% confidence intervals of association of the aPTT-increasing alleles are shown. Detailed association results are provided in the Supplementary Data 1

Results – non-examples of pleiotropy?

rs2274273 near *LGALS3*

- associated with Parkinson's disease ($OR_{23andMe}$ = 0.94, $P_{23andMe}$ = 1E−4)
- rs2274273 is a pQTL for plasma levels of galectin-3
- Co-localization analyses suggest rs2274273 isn't a causal SNP for Parkinson's

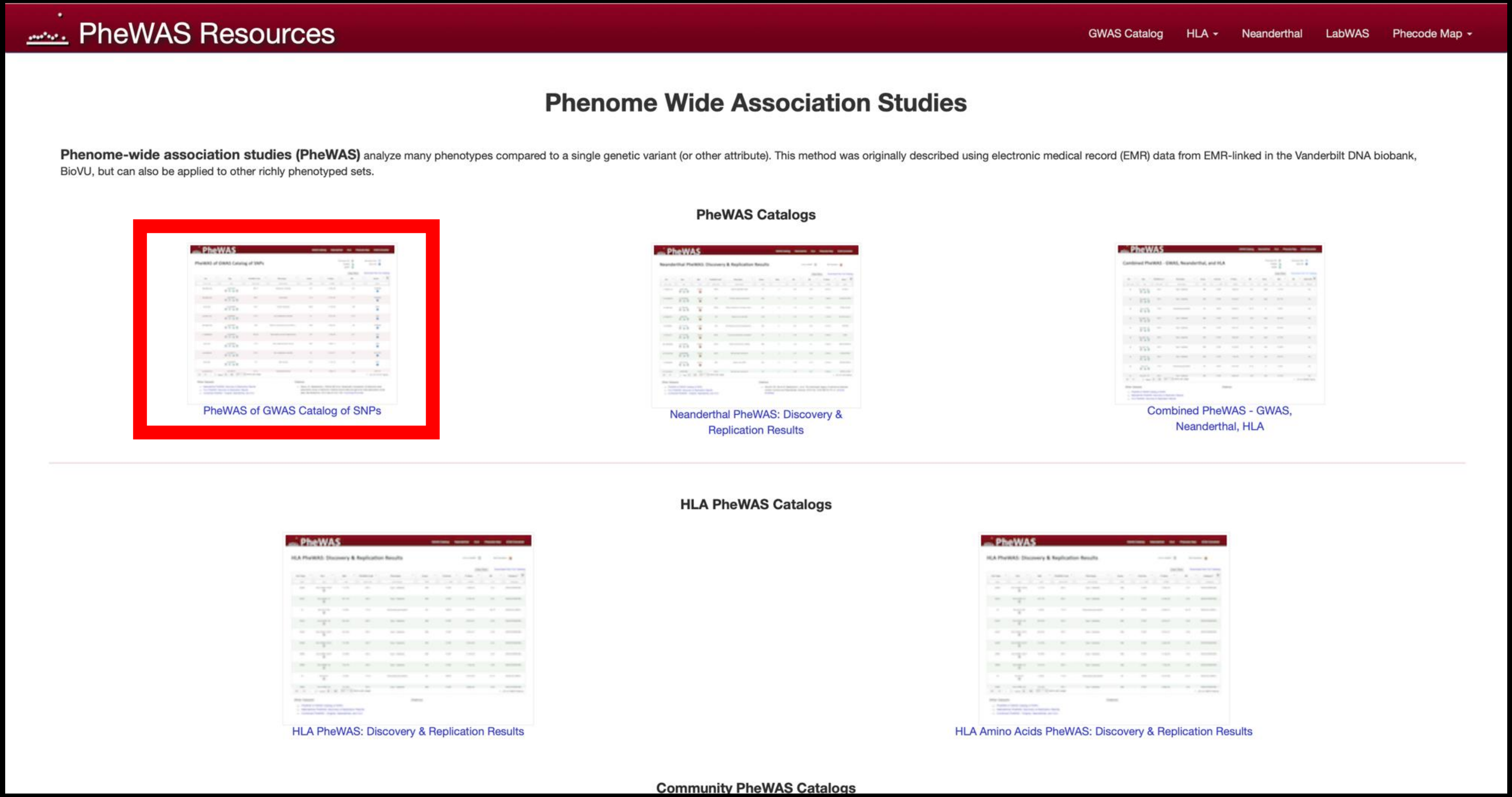
rs17724992 near GDF15

- Associated with cardiovascular-related phenotypes here
- Likely mediated by BMI (a known association for this SNP)
- Large studies haven't found associations between this SNP and coronary artery disease or blood pressure

pheWAS catalog (small, unofficial" database)

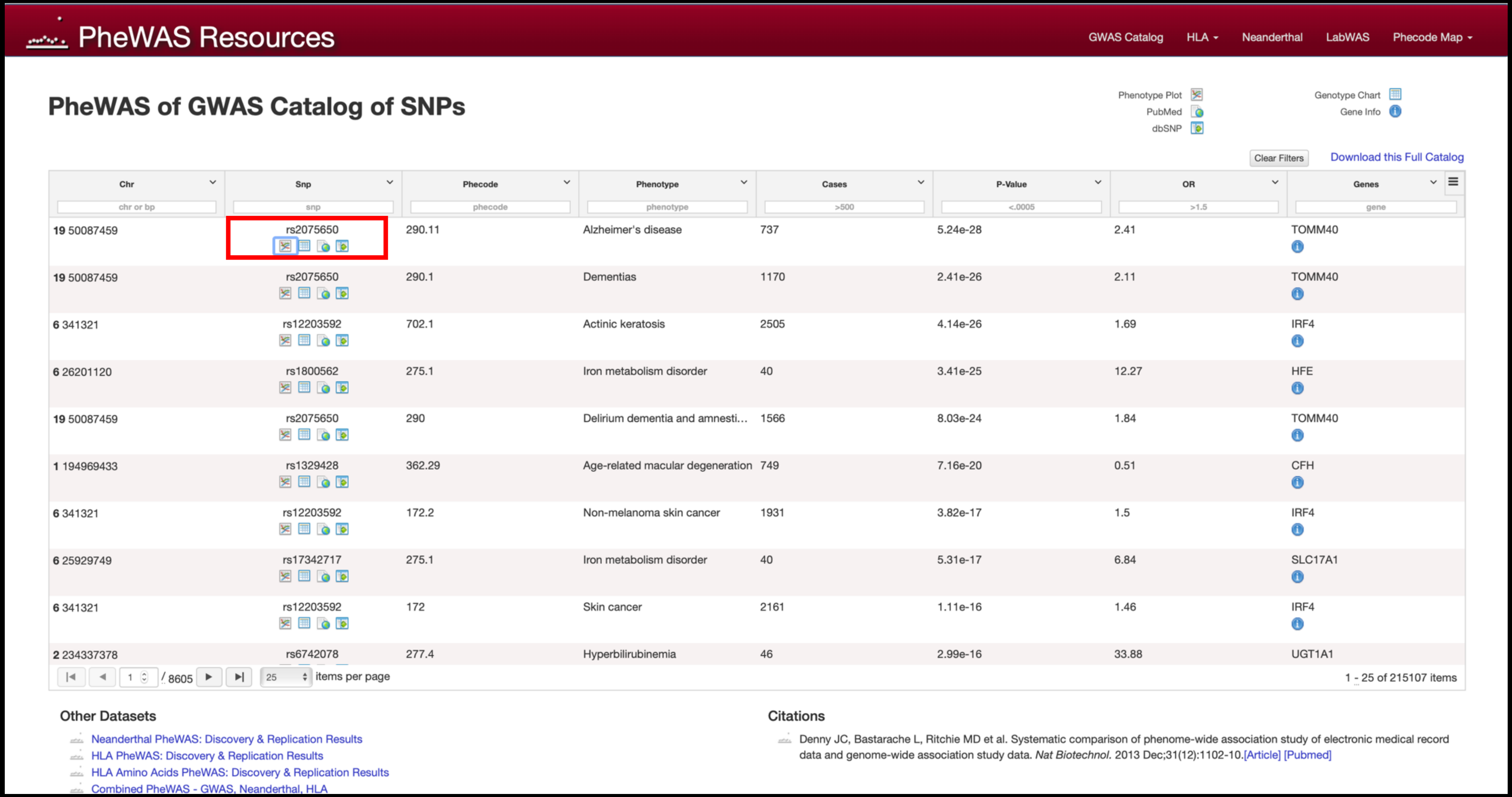
https://phewascatalog.org

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3969265/



Denny, J. C., Bastarache, L., Ritchie, M. D., Carroll, R. J., Zink, R., Mosley, J. D., Field, J. R., Pulley, J. M., Ramirez, A. H., Bowton, E., Basford, M. A., Carrell, D. S., Peissig, P. L., Kho, A. N., Pacheco, J. A., Rasmussen, L. V., Crosslin, D. R., Crane, P. K., Pathak, J., Bielinski, S. J., … Roden, D. M. (2013). Systematic comparison of phenome-wide association study of electronic medical record data and genome-wide association study data. *Nature biotechnology*, *31*(12), 1102–1110. https://doi.org/10.1038/nbt.2749

pheWAS catalog



PheWAS Resources

GWAS Catalog | HLA | Neanderthal | LabWAS | Phecode Map

PheWAS of GWAS Catalog of SNPs

Phenotype Plot | Genotype Chart | PubMed | Gene Info | dbSNP

Clear Filters | Download this Full Catalog

Chr	Snp	Phecode	Phenotype	Cases	P-Value	OR	Genes
chr or bp	snp	phecode	phenotype	>500	<.0005	>1.5	gene
19 50087459	rs2075650	290.11	Alzheimer's disease	737	5.24e-28	2.41	TOMM40
19 50087459	rs2075650	290.1	Dementias	1170	2.41e-26	2.11	TOMM40
6 341321	rs12203592	702.1	Actinic keratosis	2505	4.14e-26	1.69	IRF4
6 26201120	rs1800562	275.1	Iron metabolism disorder	40	3.41e-25	12.27	HFE
19 50087459	rs2075650	290	Delirium dementia and amnesti…	1566	8.03e-24	1.84	TOMM40
1 194969433	rs1329428	362.29	Age-related macular degeneration	749	7.16e-20	0.51	CFH
6 341321	rs12203592	172.2	Non-melanoma skin cancer	1931	3.82e-17	1.5	IRF4
6 25929749	rs17342717	275.1	Iron metabolism disorder	40	5.31e-17	6.84	SLC17A1
6 341321	rs12203592	172	Skin cancer	2161	1.11e-16	1.46	IRF4
2 234337378	rs6742078	277.4	Hyperbilirubinemia	46	2.99e-16	33.88	UGT1A1

1 / 8605 | 25 items per page | 1 - 25 of 215107 items

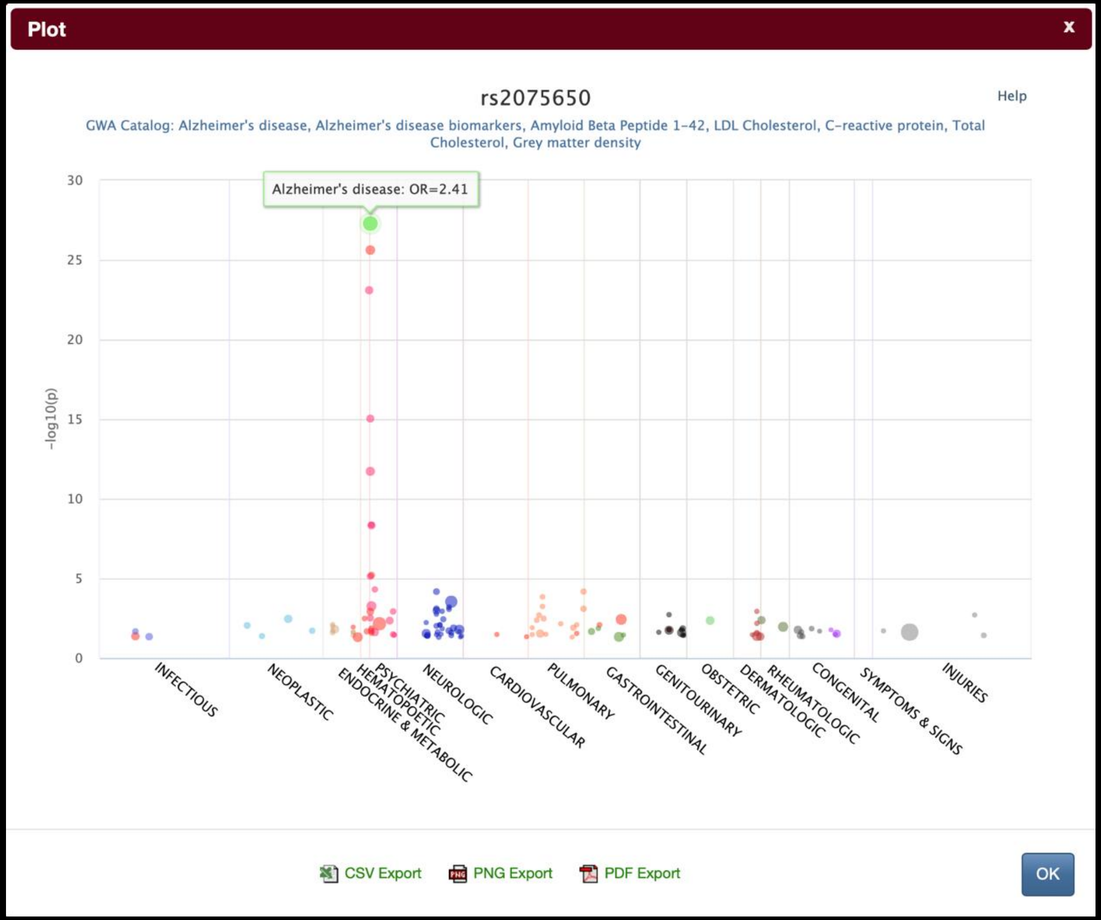
Other Datasets

- Neanderthal PheWAS: Discovery & Replication Results
- HLA PheWAS: Discovery & Replication Results
- HLA Amino Acids PheWAS: Discovery & Replication Results
- Combined PheWAS - GWAS, Neanderthal, HLA

Citations

- Denny JC, Bastarache L, Ritchie MD et al. Systematic comparison of phenome-wide association study of electronic medical record data and genome-wide association study data. *Nat Biotechnol.* 2013 Dec;31(12):1102-10.[Article] [Pubmed]

pheWAS catalog



pheWAS catalog

Genotype x

rs2075650

Alzheimer's disease

Genotype Frequencies

	GG	GA	AA
Cases 737	43 (6%)	256 (35%)	438 (59%)
Controls 9456	179 (2%)	2246 (24%)	7031 (74%)

Allele Frequencies

	G	A
Cases 737	342 (23%)	1132 (77%)
Controls 9456	2604 (14%)	16308 (86%)

OK

pheWAS catalog

Sort by SNP/GWAS attribute

Search/filter (not all of these seem to work right)

PheWeb

A system for plotting pheWAS results.

https://pheweb.sph.umich.edu

Some pheWAS with ~publicly available stats

Access is spotty…

- AllofUs (https://allbyall.researchallofus.org)
- VA Million Veteran Program (https://www.mvp.va.gov/pwa/discover-mvp-data)
- AstraZeneca PheWAS Portal (https://azphewas.com)
- Other UKBB studies
- FinnGEN (https://docs.finngen.fi/working-outside-the-sandbox/link-to-how-to-use-pheweb)

Other things to do with summary stats of multiple traits

LD score regression

- Distinguish polygenicity from confounding
  - Input: GWAS summary stats
  - Output: intercept (confounding) + slope (heritability)
- Estimate/partition heritability
  - Input: GWAS summary stats
  - Output: heritability, and fractions of heritability by annotation
- Genetic correlations
  - Input: GWAS summary stats for two traits
  - Output: r_g (correlation of SNPs across traits; how similar the genetic architectures are)

Other things to do with summary stats of multiple traits

Mendelian randomization

- Logic
  - If a SNP changes the outcome only because it changes the exposure, then…
  - …effect of SNP on outcome ÷ effect of SNP on exposure = causal effect of exposure on outcome
- Use genetic variant to ask, “Does this exposure *cause* this trait?”
- Assumptions
  - You already know some particular SNP is associated with the exposure (relevance)
  - The SNP is *not* associated with any confounders (independence)
  - The SNP affects the trait *only* through the exposure
  - These need to be checked - but checking can be difficult in practice!
- Output
  - Evidence consistent with causality
  - Estimated causal effect (if assumptions hold)

MR (continued)

Example

- Higher LDL cholesterol is associated with coronary artery disease risk
- Is this causal?
  - Use MR
  - Instruments: SNPs (alleles) that lower LDL
  - Alleles that lower LDL risk also lower CAD risk
  - → consistent with LDL causally increasing CAD risk

Summary MR

You can do MR without individual level data.

- Use summary stats from two GWAS (same SNPs, aligned alleles)
  - GWAS 1: exposure
  - GWAS 2: outcome of interest
- Summary MR
  - Use significant SNPs from the exposure GWAS
  - Combines SNP effects via β(outcome) / β(exposure)
  - Outputs estimate of causal effect of exposure on outcome
- Same assumptions as before, with more caveats
  - Weak instrument bias: need SNPs that strongly affect exposure
  - Harder to check assumptions without individual-level data
  - Specifically, ruling out horizontal pleiotropy is harder